

Deep Learning for Computer Vision

Dr. Nasser M. Nasrabadi

Virtual Lectures, Wednesdays/Thursdays, 7.20-10pm

Nasser.Nasrabadi@mail.wvu.edu

Siamese + Coupled Neural Networks Architecture

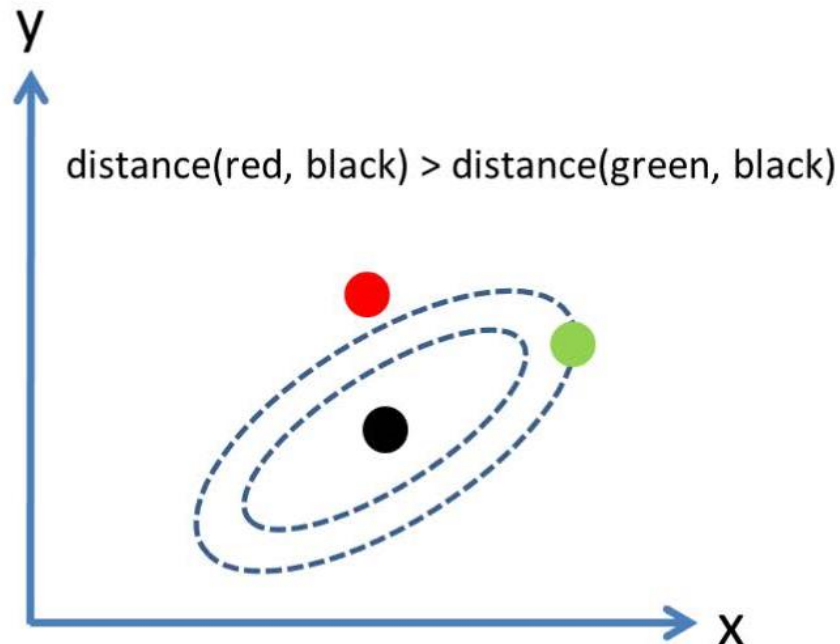
Siamese Deep Neural Networks & Coupled Deep Neural Networks

1. Sumit Chopra, Raia Hadsell, Yann LeCun, “[Learning a Similarity Metric Discriminatively, with Application to Face Verification](#),” CVPR July 2005.
2. Raia Hadsell, Sumit Chopra, Yann LeCun. “[Dimensionality Reduction by Learning an Invariant Mapping](#),” CVPR '06 Proceedings of the 2006 IEEE Computer Society Conference on Computer Vision and Pattern Recognition - Volume 2, Pages 1735-1742.
3. Joplin Hu, Jiwen Lu, Yap-Peng, “[Discriminative Deep Metric Learning for Face Verification in the Wild](#),” CVPR 2014
4. Benjamin S. Riggan, Christopher Reale And Nasser M. Nasrabadi, “[Coupled Auto-Associative Neural Network for Heterogeneous Face Recognition](#),” IEEE Access Sept. 30 2015.

Similarity Metric Learning

The performance of many learning and datamining algorithms depend critically on their being given a good metric over the input space. For instance, K-means, nearest-neighbors classifiers and kernel algorithms such as SVMs all need to be given good metrics that reflect reasonably well the important relationships between the data.

- Euclidean distance vs Mahalanobis distance



Euclidean Distance:

$$D(\mathbf{x} - \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \{(\mathbf{x} - \mathbf{y})^T (\mathbf{x} - \mathbf{y})\}^{-2}$$

Mahalanobis Distance:

$$D(\mathbf{x}, \mathbf{y}) = \{(\mathbf{x} - \mathbf{y})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{y})\}^{-2}$$

$$\begin{aligned} \text{Cov}(A) &= \begin{bmatrix} \frac{\sum (x_i - \bar{X})(x_i - \bar{X})}{N} & \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{N} \\ \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{N} & \frac{\sum (y_i - \bar{Y})(y_i - \bar{Y})}{N} \end{bmatrix} \\ &= \begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(Y, X) \\ \text{Cov}(X, Y) & \text{Cov}(Y, Y) \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} \text{Var}(x_1) & \dots & \text{Cov}(x_n, x_1) \\ \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \dots & \text{Var}(x_n) \end{bmatrix}$$

Types of Metrics

Broadly distance metrics are of two kinds:

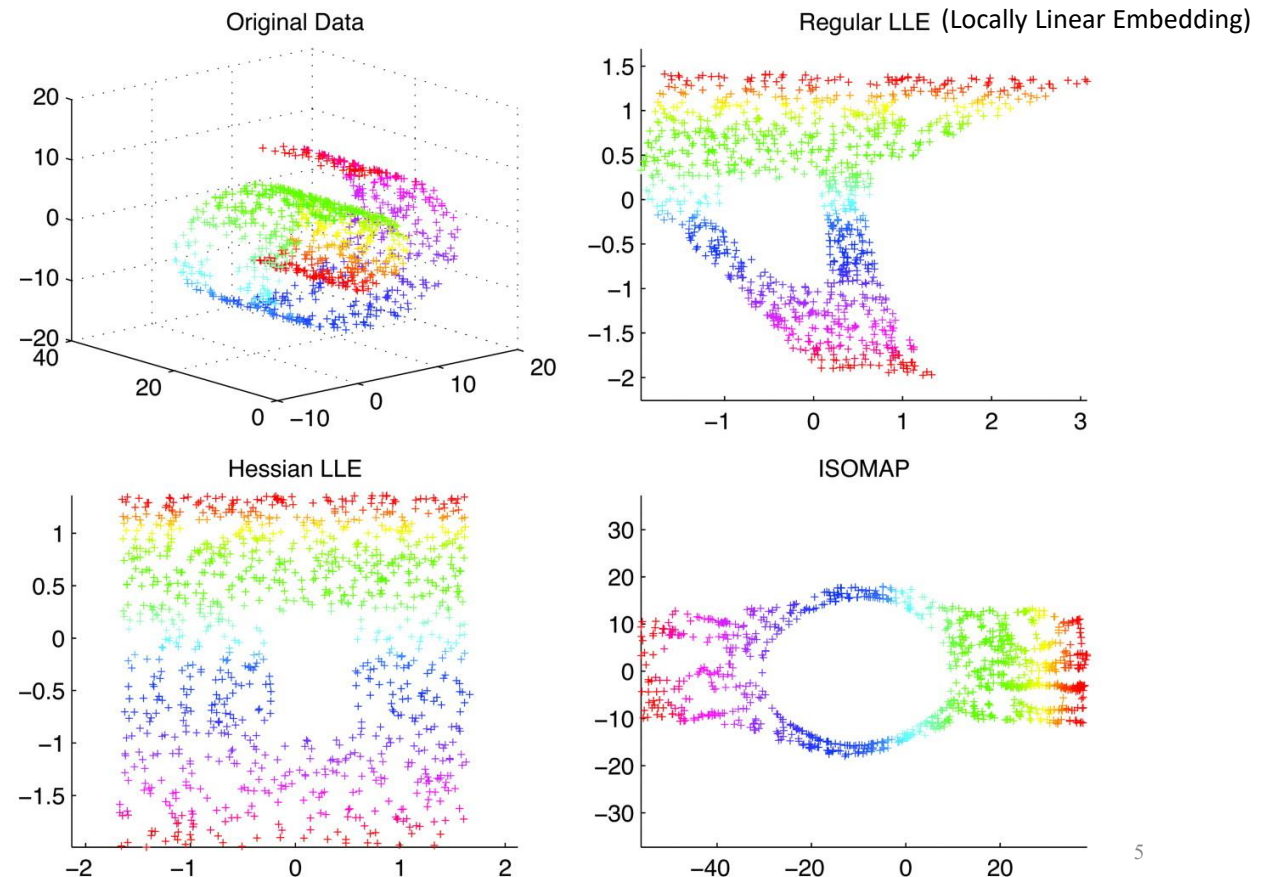
- Pre-defined Metrics: Metrics which are fully specified without the knowledge of data. e.g., Euclidian Distance: $D(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^T(\mathbf{x} - \mathbf{y})$
- Learned Metrics: Metrics which can only be defined with the knowledge of the data.
e.g., Mahalanobis Distance: $D(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^T \mathbf{M} (\mathbf{x} - \mathbf{y})$; where $\mathbf{M}$ is an embedding matrix (Covariance Matrix= $\mathbf{C}^{-1}$) that is estimated from the data.

Learned Metrics are of two types:

- Unsupervised : Use unlabeled data
- Supervised: Use Labeled data

Metric Learning - Data Embedding – dimensionality reduction

- **PCA & MDS:** Two popular forms of **linear** dimensionality reduction are the methods of principal component analysis (PCA) and multidimensional scaling (MDS). Both PCA and MDS are eigenvector methods designed to model linear variabilities in high dimensional data. In PCA, one computes the linear projections of greatest variance from the top eigenvectors of the data covariance matrix.
- **Manifold learning** is an approach to **non-linear dimensionality reduction**. Algorithms for this task are based on the idea that the dimensionality of many data sets is only artificially high.
- Siamese network does a non-linear embedding that preserve the data similarity or structure.

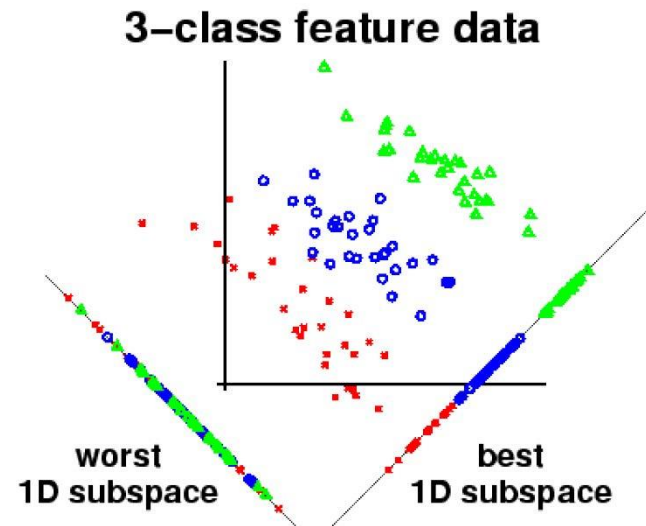


An Example of Unsupervised Metrics Learning (Linear Discriminant Analysis (LDA))

In Fisher-LDA*, the goal is to project the data to a space such that the ratio of “between class covariance ($\mathbf{S}_B$)” to “within class covariance ($\mathbf{S}_w$)” is maximized.

This is given by:

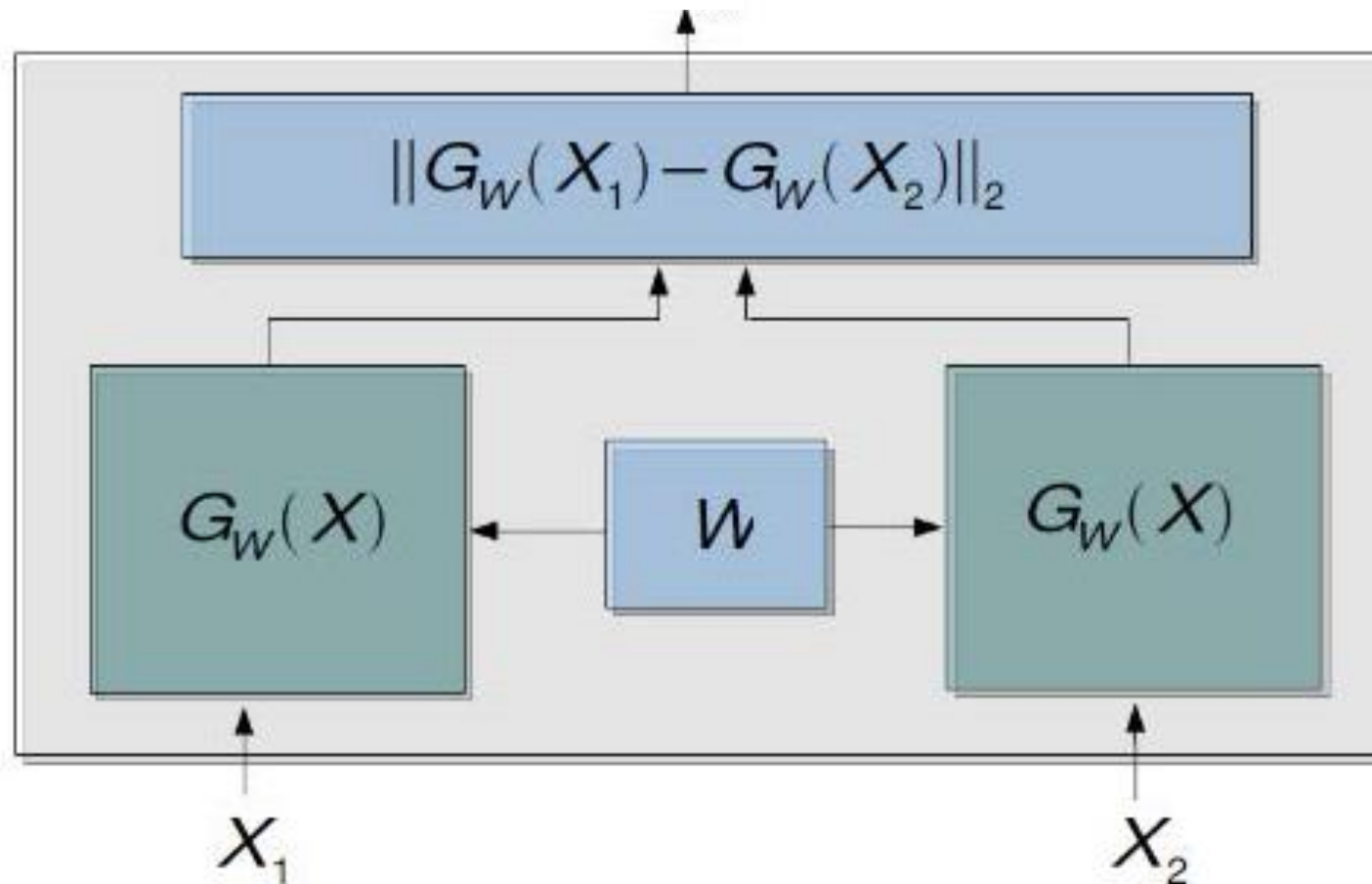
$$J(\mathbf{w}) = \underset{\mathbf{w}}{Max} \left\{ \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}} \right\}$$



Siamese Architecture (A Non-Linear Metric Learning)

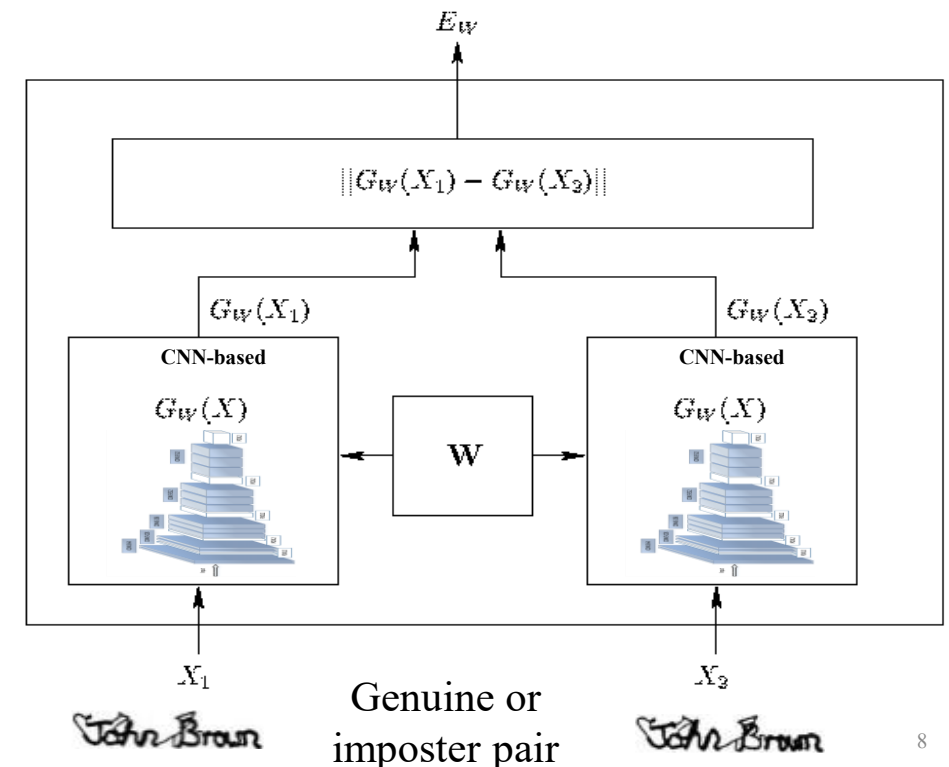
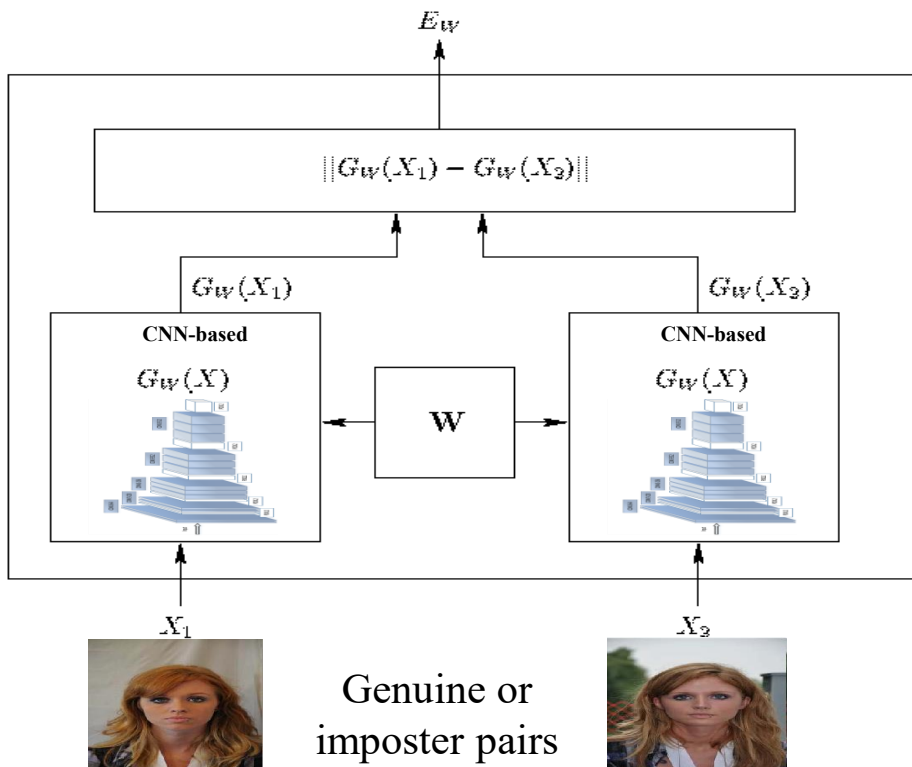
- Consider the Euclidean distance measure:

$$D_W(\mathbf{x}_1, \mathbf{x}_2) = \|G_w(\mathbf{x}_1) - G_w(\mathbf{x}_2)\|_2$$



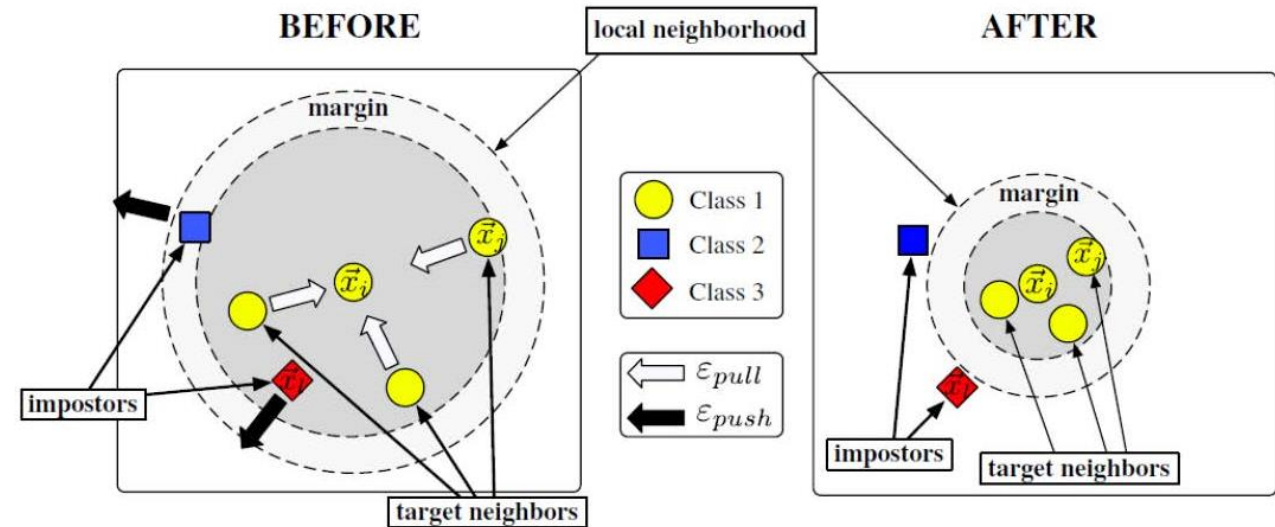
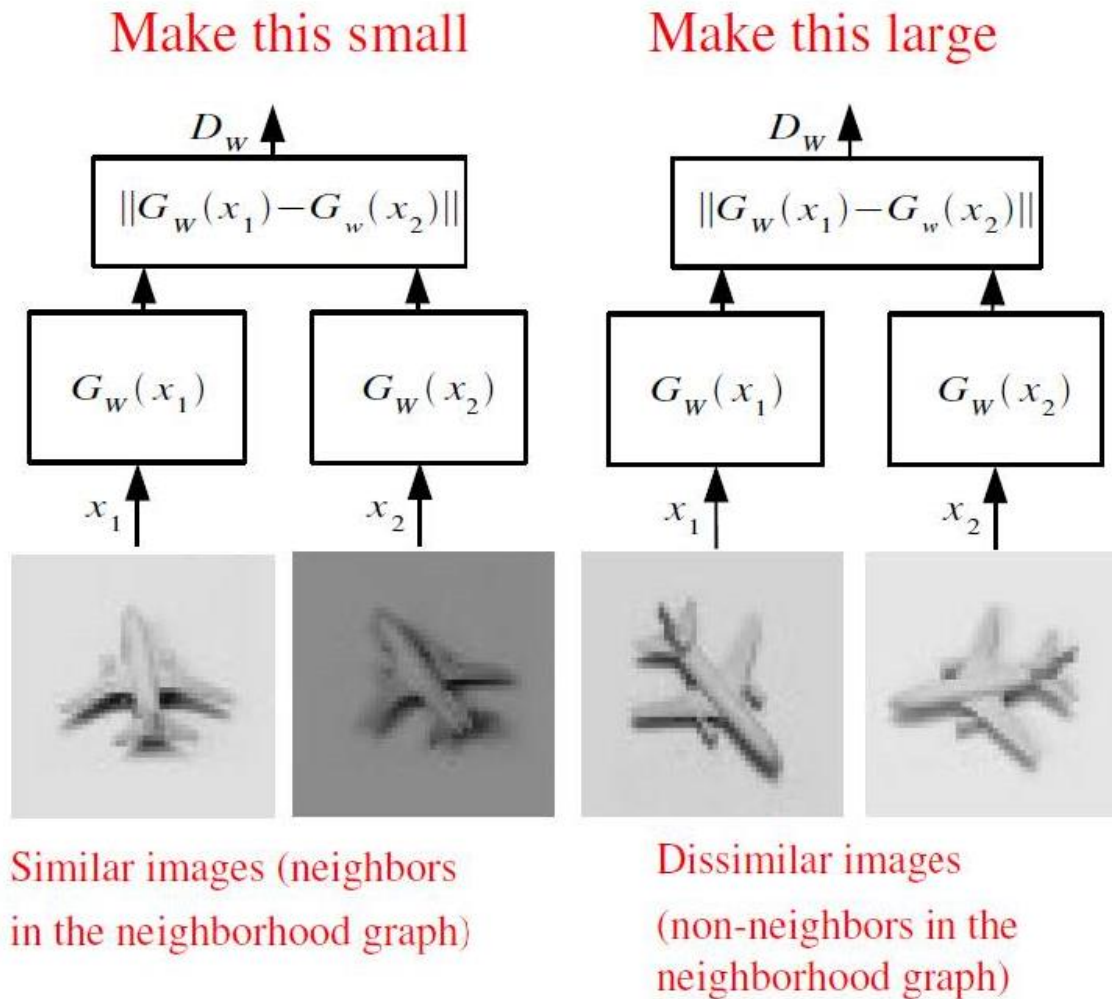
Siamese Network Non-Linear Similarity Metric Learning

- Siamese network is designed for *similarity metric* learning
- The idea is to learn a function that maps the input patterns into a target space such that pairs of photos (feature vectors) belonging to the same person will become more closer (distance-wise) compared to pairs of photos (feature vectors) from different persons.
- It can also be used as dimensionality reduction by learning an invariant mapping (similar images in mapping domain are also close to each other. **This is called feature embedding.**
- For example these identical twin are the same? Are these the same signatures?



Siamese Architecture and Idea of Contrastive Loss Function

- Similar images (genuine pairs) should cluster in the embedded domain and dissimilar images (imposter pairs) should diverge from each other in the embedded domain



Siamese Metric Distance and Contrastive Cost Functions

- The two CNN networks are the same i.e., have the same weights (weight sharing)
- Define an L_2 -norm distance (Euclidean): $D_W(\mathbf{x}_i, \mathbf{x}_j) = \|G_w(\mathbf{x}_i) - G_w(\mathbf{x}_j)\|_2$
- Trained on a pair of samples from the training data
- If $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same category i.e., $y = 0$ we want $D_W(\mathbf{x}_i, \mathbf{x}_j)$ to be small.
- If $\mathbf{x}_1$ and $\mathbf{x}_2$ belong to different categories i.e., $y = 1$ we want $D_W(\mathbf{x}_1, \mathbf{x}_2)$ to be large.
- Using above conditions, we define a **contrastive loss** for an input pair $\mathbf{x}_i$ and $\mathbf{x}_j$ as

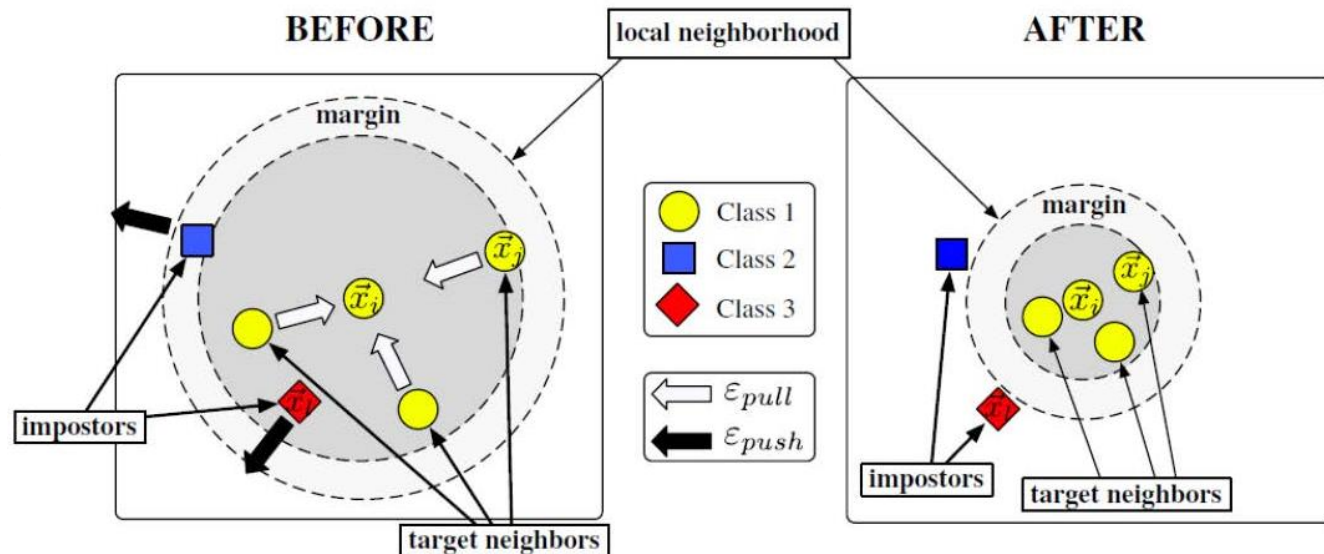
$$L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = (1 - y)L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) + yL_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j))$$

- Where for genuine pairs:

$$L_{gen}(D_W(\mathbf{x}_1, \mathbf{x}_2)) = \frac{1}{2} D_W(\mathbf{x}_1, \mathbf{x}_2)^2 = \frac{1}{2} \|G_w(\mathbf{x}_1) - G_w(\mathbf{x}_2)\|_2^2$$

- For imposter pairs:

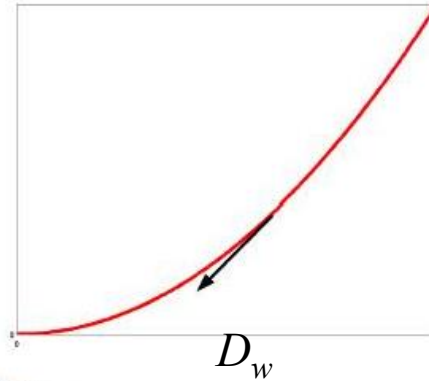
$$L_{imp}(D_W(\mathbf{x}_1, \mathbf{x}_2)) = \frac{1}{2} \max(0, (m - D_W(\mathbf{x}_1, \mathbf{x}_2)))^2$$



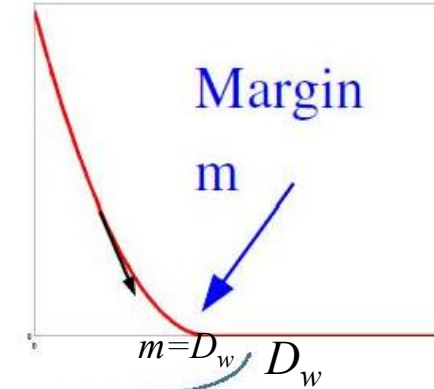
Siamese Contrastive Loss Function

Loss function

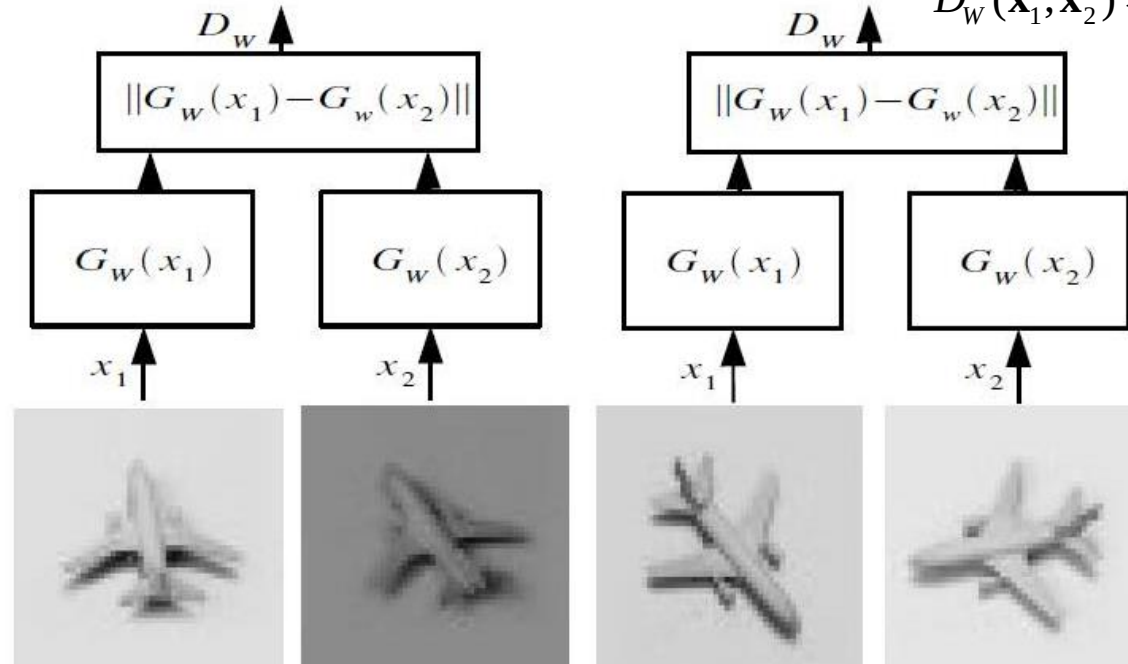
$$L_{\text{similar}} = \frac{1}{2} D_w^2$$



$$L_{\text{dissimilar}} = \frac{1}{2} \{ \max(0, m - D_w) \}^2$$



Hinge Loss



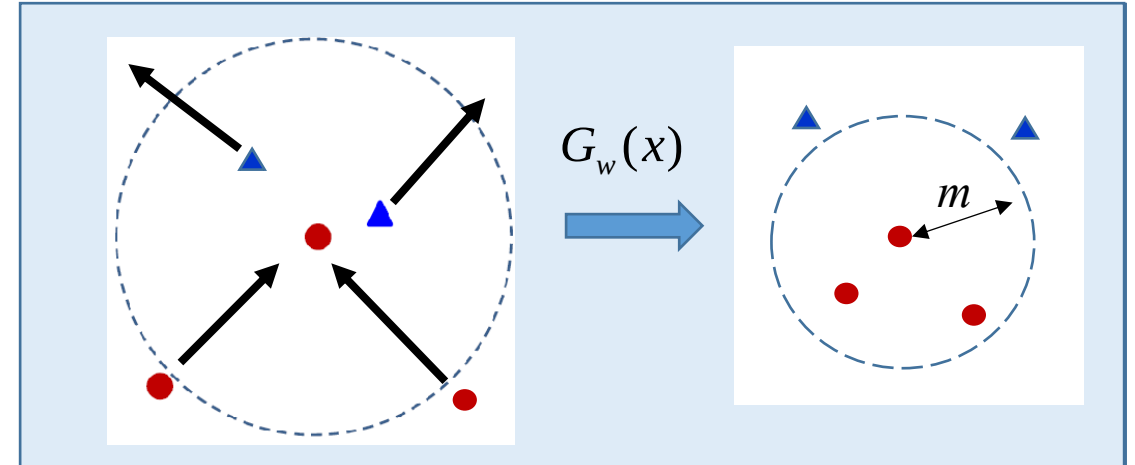
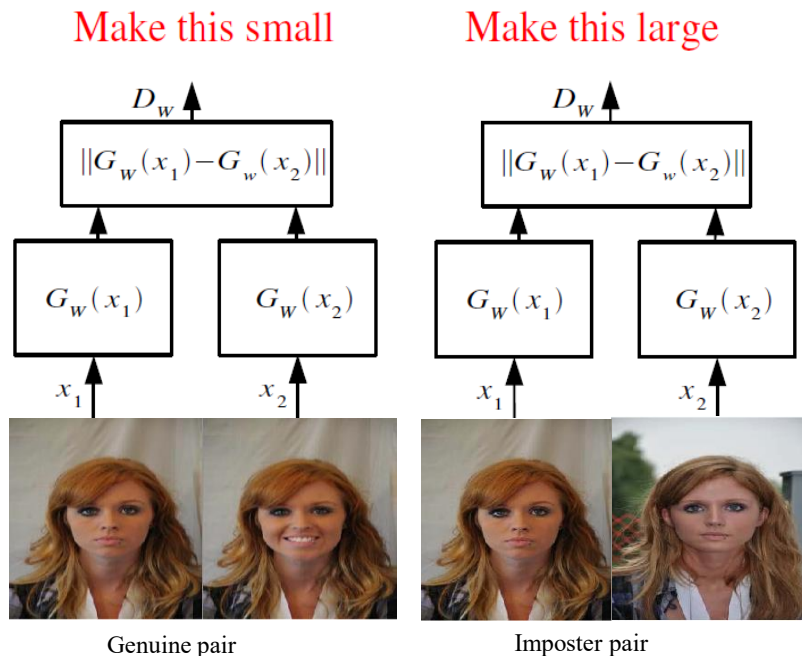
$$D_w(\mathbf{x}_1, \mathbf{x}_2) = \|G_w(\mathbf{x}_1) - G_w(\mathbf{x}_2)\|_2$$

Siamese Contrastive Loss

- Define an L_2 -norm distance (Euclidean): $D_W(\mathbf{x}_i, \mathbf{x}_j) = \|G_w(\mathbf{x}_i) - G_w(\mathbf{x}_j)\|_2$
- For Genuine Pairs ($y=0$): $L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} D_W(\mathbf{x}_i, \mathbf{x}_j)^2 = \frac{1}{2} \|G_w(\mathbf{x}_i) - G_w(\mathbf{x}_j)\|_2^2$
- For Imposter pairs ($y=1$): $L_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \max(0, m - D_W(\mathbf{x}_i, \mathbf{x}_j))^2$
- m is a margin (anything inside the margin are genuine pairs and outside the margin are imposters).
- We have to minimize the total cost for N samples

$$\ell(W) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N L(W, (y, \mathbf{x}_i, \mathbf{x}_j))$$

$$L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = (1 - y) L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) + y L_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j))$$



Details of Contrastive Loss (Push and Pull)

- Consider minimizing the cost function with respect to W for one input pair

- We need to differentiate
$$\frac{\partial}{\partial W} L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \frac{\partial}{\partial W} ((1-y)L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) + yL_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j)))$$

- Consider a genuine pair thus $y = 0$,
$$\frac{\partial}{\partial W} L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \frac{\partial}{\partial W} L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = + D_W \frac{\partial D_W}{\partial W}$$

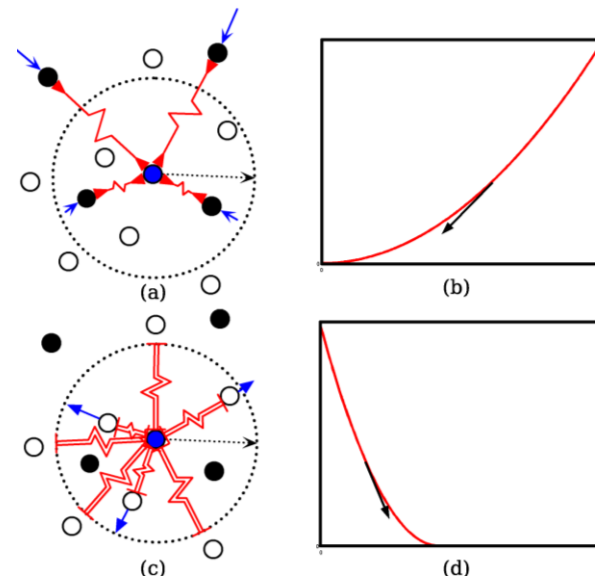
- Consider a imposter pair thus $y = 1$,

$$\frac{\partial}{\partial W} L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \frac{\partial}{\partial W} L_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \frac{\partial}{\partial W} \max(0, (m - D_W(\mathbf{x}_i, \mathbf{x}_j)))^2$$

$$\frac{1}{2} \frac{\partial}{\partial W} L_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = - [(m - D_W(\mathbf{x}_i, \mathbf{x}_j))] \frac{\partial D_W}{\partial W}$$

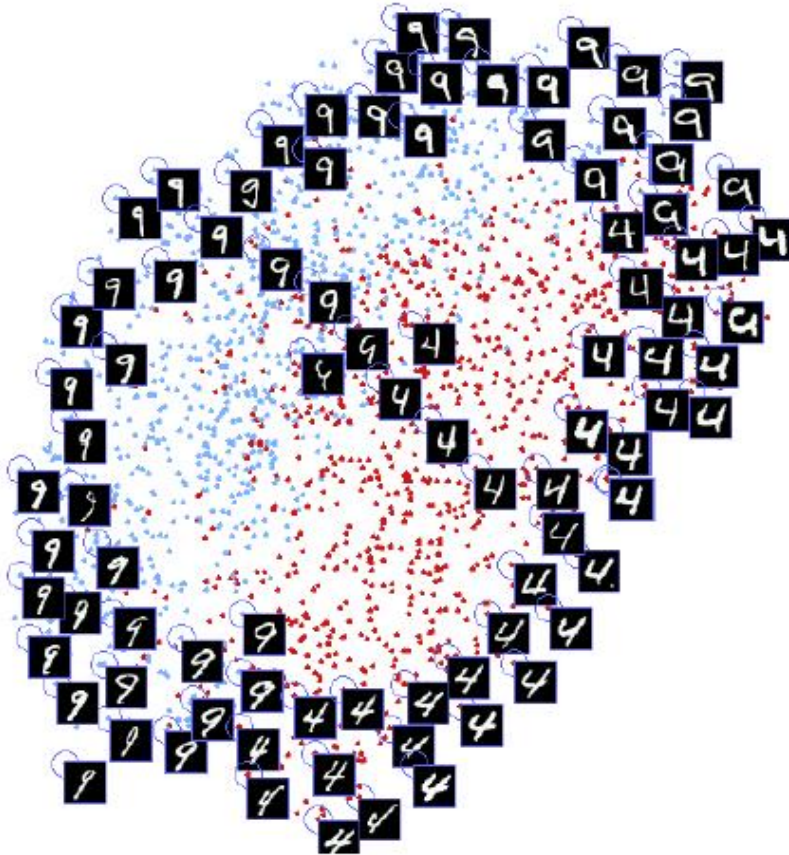
Positive means pulling together in the mapped domain

Negative means pushing away in the mapped domain

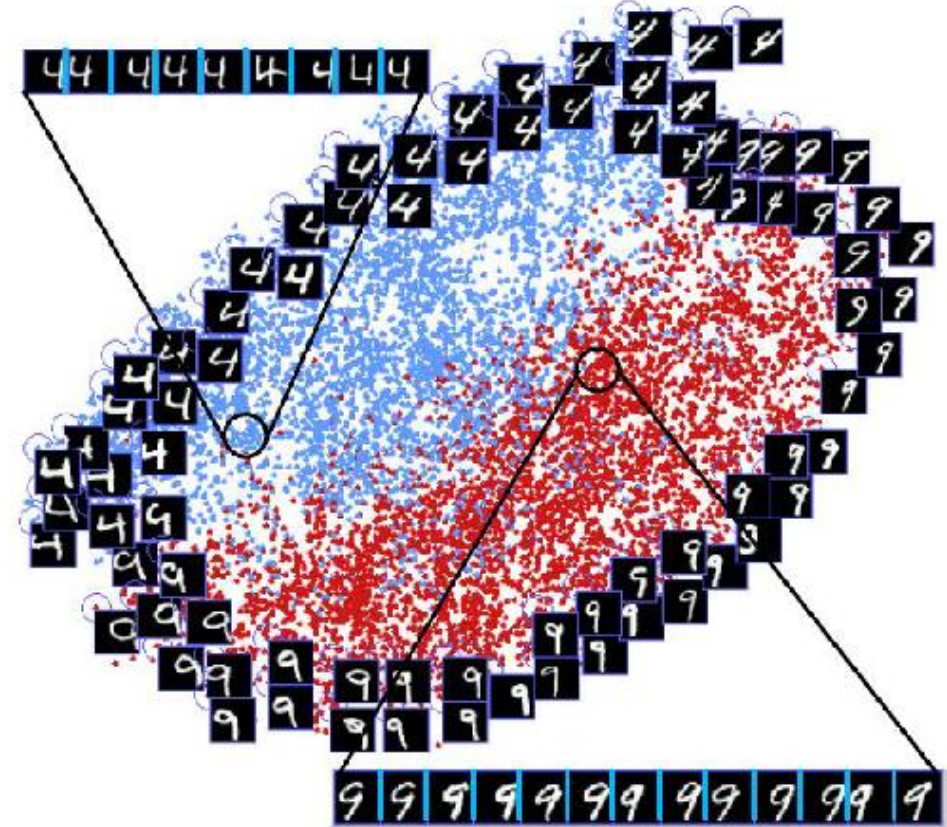


Siamese Networks – Application in Dimensionality Reduction

- Consider digits 4 and 9 from the MNIST dataset. Same digits are mapped close to each other:



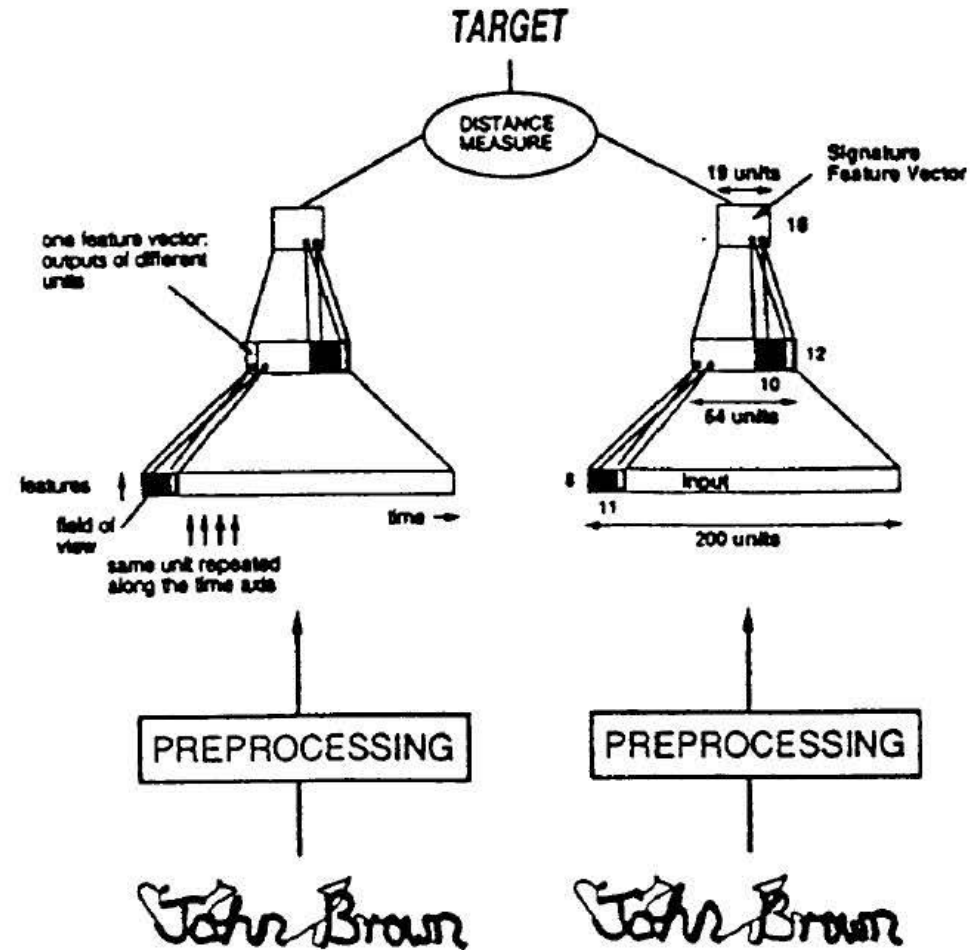
Learned Mapping of MNIST samples



Learning a Shift Invariant Mapping of MNIST samples

Siamese Networks – Application in Signature Verification

- The input is 8(feature) x 200(time) units.
- The cosine distance was used, (1 for genuine pairs, -1 for forgery pairs)

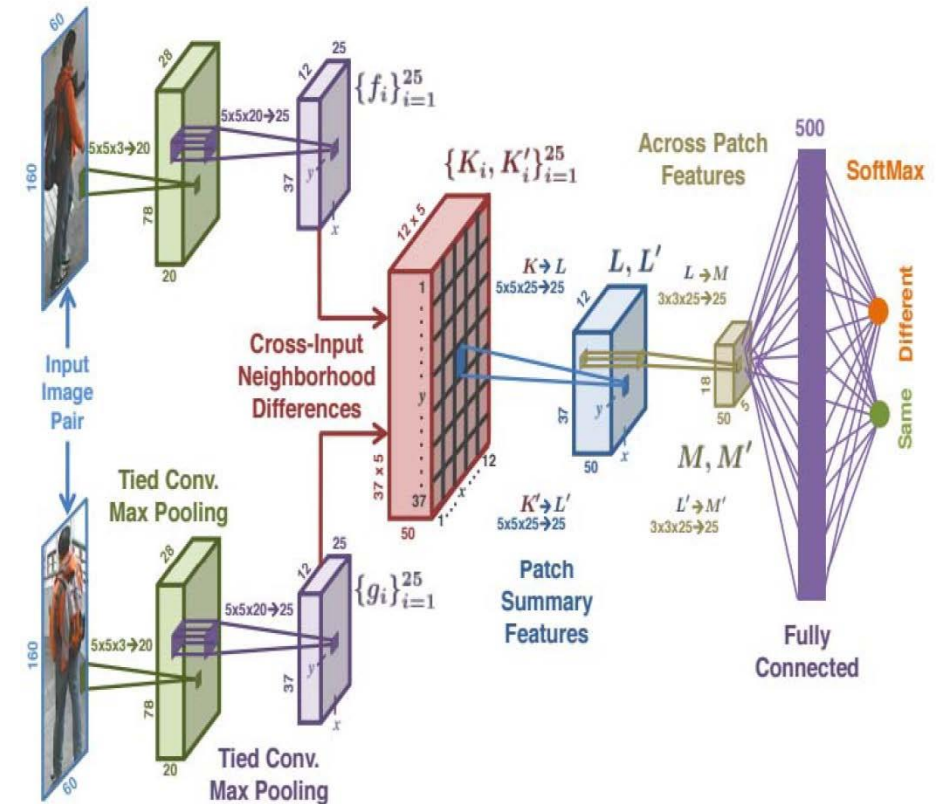


Siamese Networks – Application in Person Re-Identification

True
positive

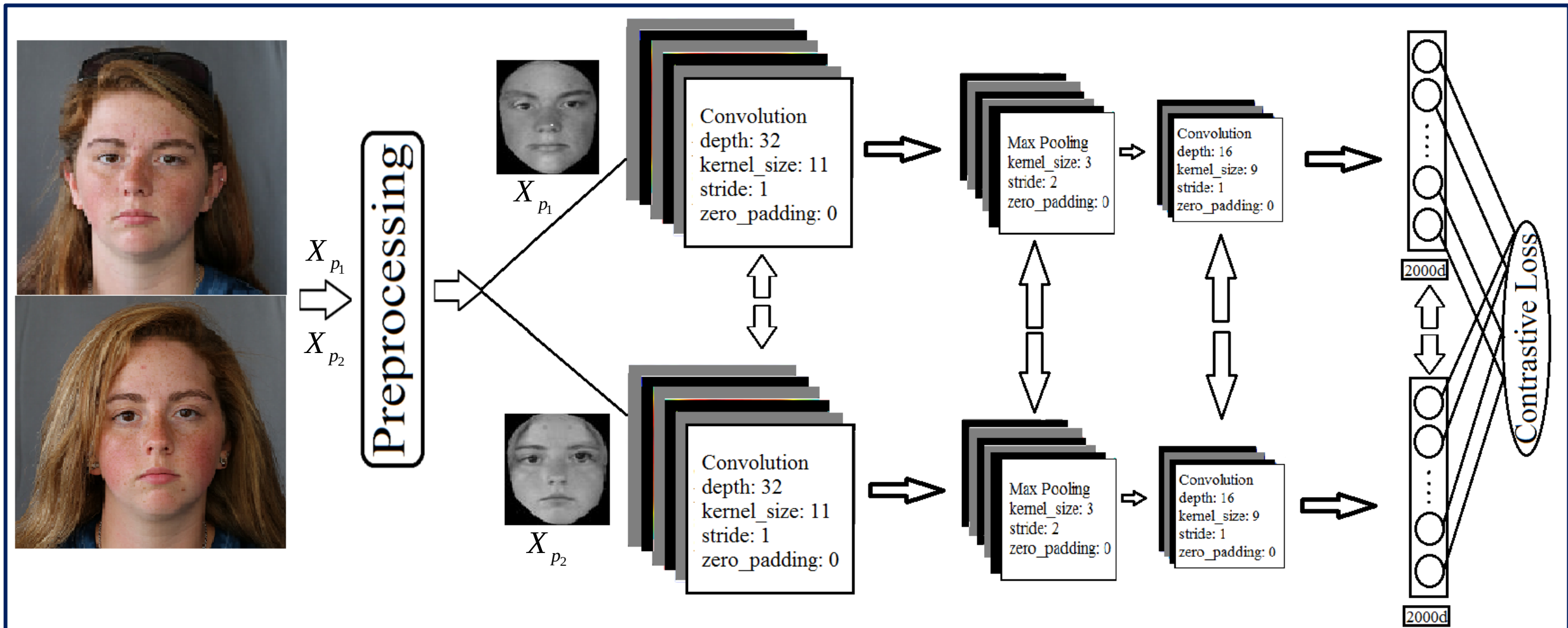


True
negative



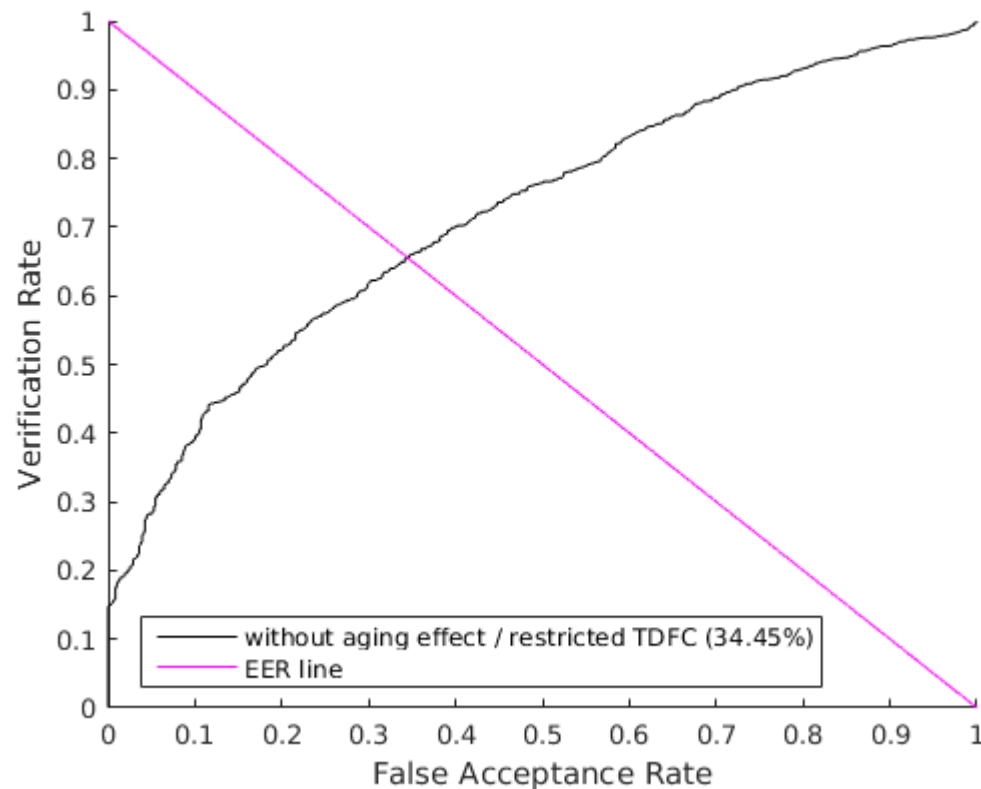
Siamese Network for Identical Twin Discrimination

- Use VGG-net 16-layer pre-trained on CASIA face imagery then trained on the twin dataset consisting of 1187 distinctive identities.
- The WVU TDFC dataset used in our experiments is gathered in the year of 2010–2014, which consists of 1187 distinctive identities. Each image is of size 4288×2848 .
- All images re-resized to 128×78 in the experiment.

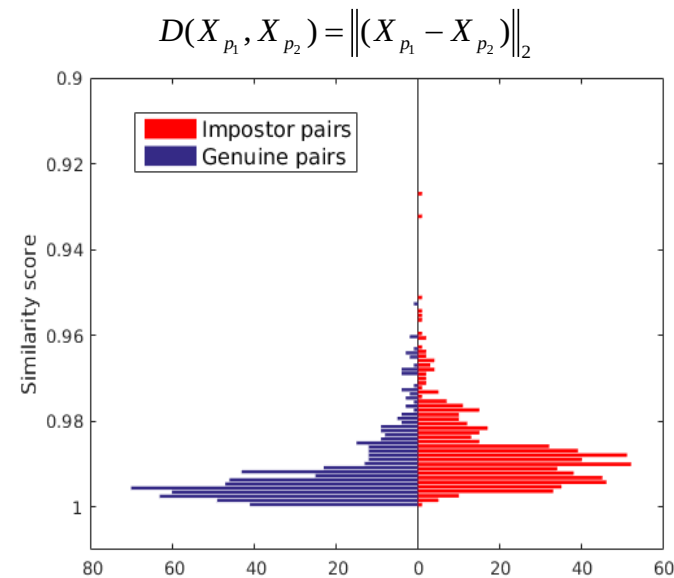


ROC Curves for Identical Twins

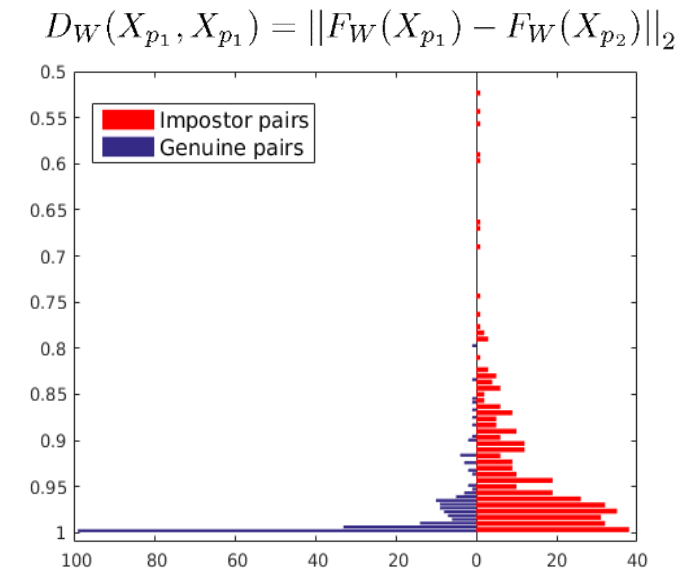
- The WVU TDFC dataset used in our experiments is gathered in the year of 2010–2014, which consists of 1187 distinctive identities. Each image is of size 4288×2848 .
- All images re-resized to 128×78 in the experiment.



ROC curve for twin face verification



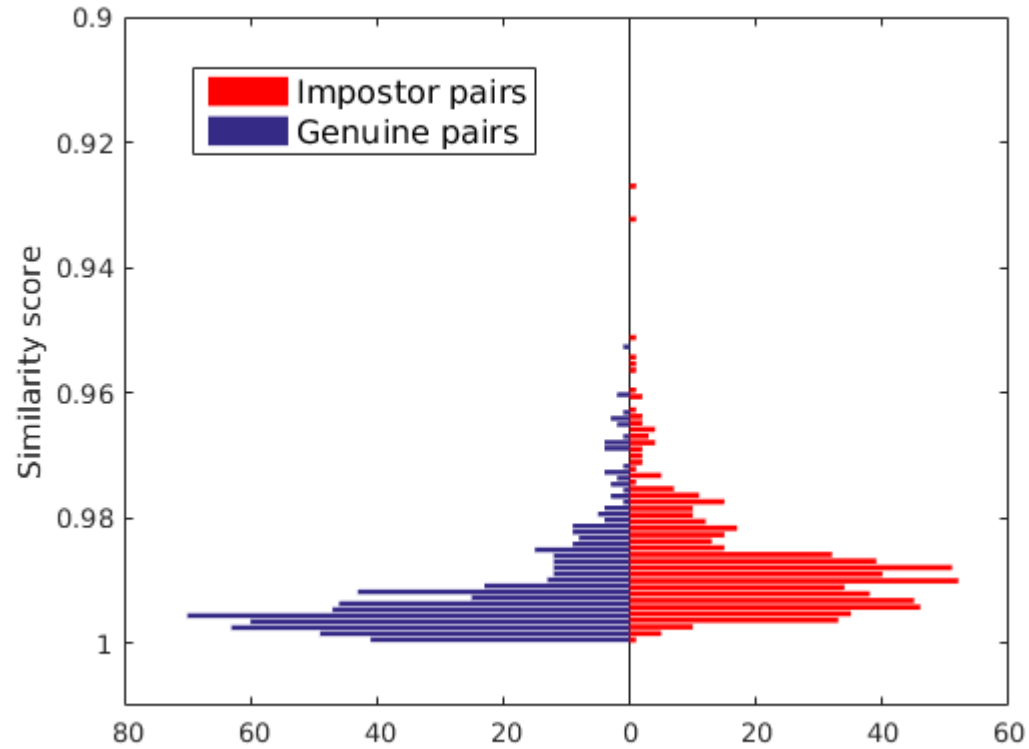
Distributions of genuine and impostor pairs by using Euclidean similarity score for the raw test samples.



Distributions of genuine and impostor pairs in the mapped feature subspace by using the Siamese CNN similarity score.

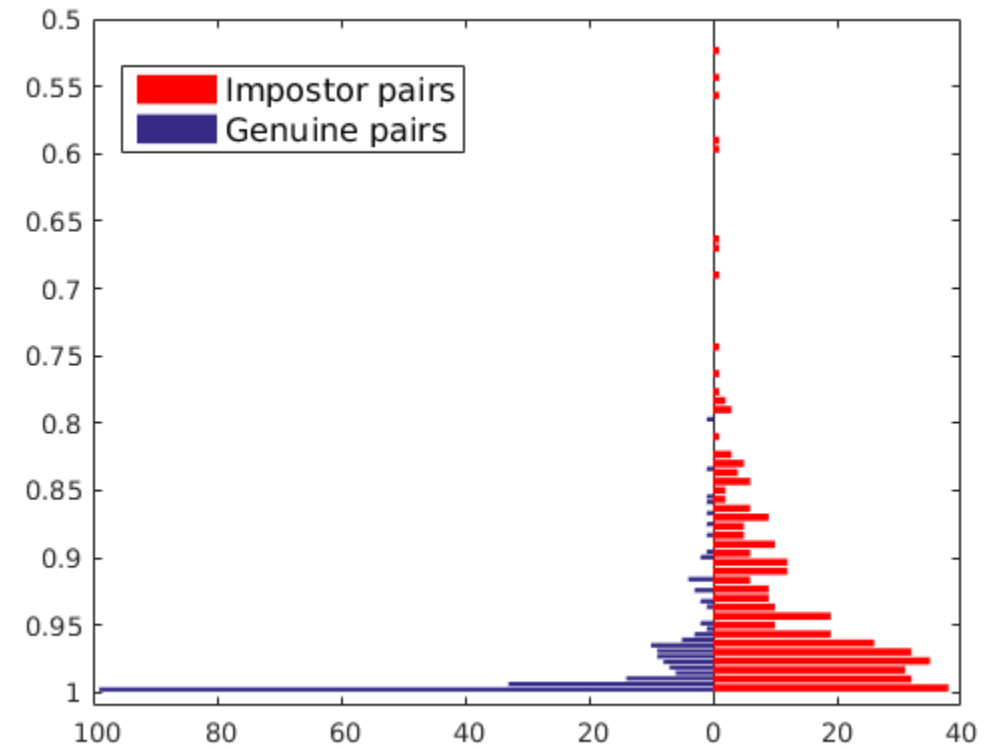
ROC Curves for Twin and Look-alike Verification

Similarity Scores for Identical Twins



Distribution of genuine and impostor pairs by using Euclidean similarity score for the test samples.

$$D(X_{p_1}, X_{p_2}) = \|(X_{p_1} - X_{p_2})\|_2$$

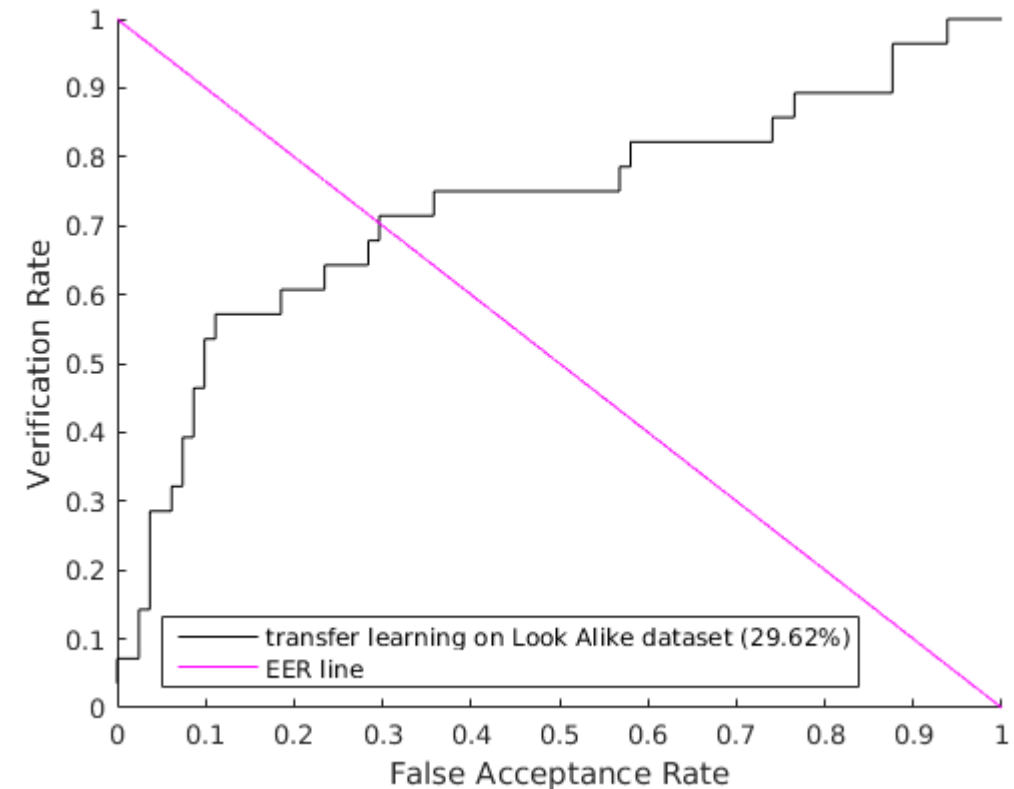
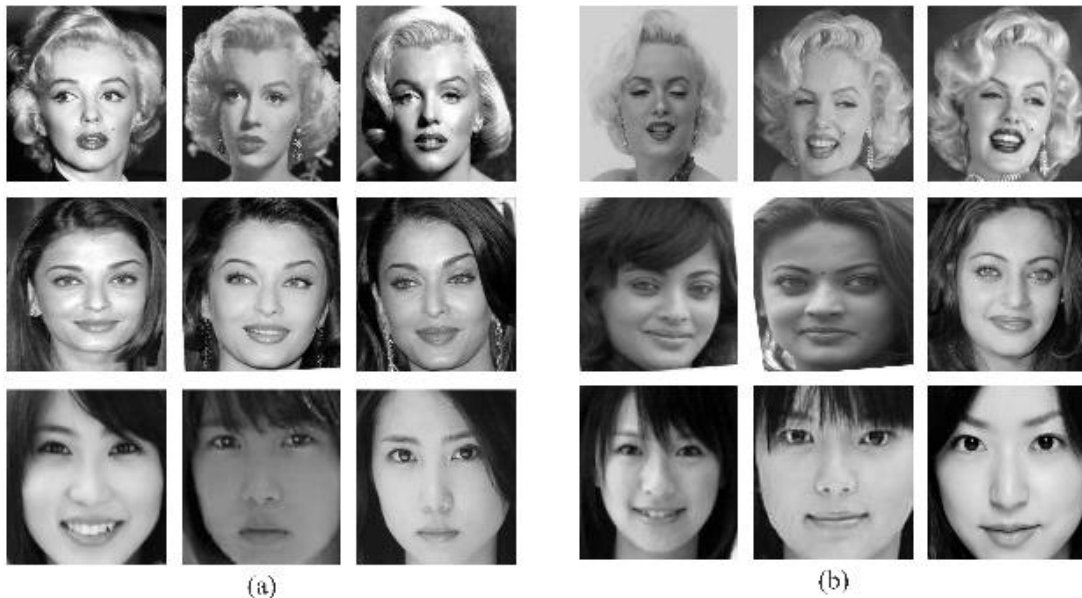


Distribution of genuine and impostor pairs in the target feature subspace by using the Siamese CNN similarity score.

$$D_W(X_{p_1}, X_{p_2}) = \|F_W(X_{p_1}) - F_W(X_{p_2})\|_2$$

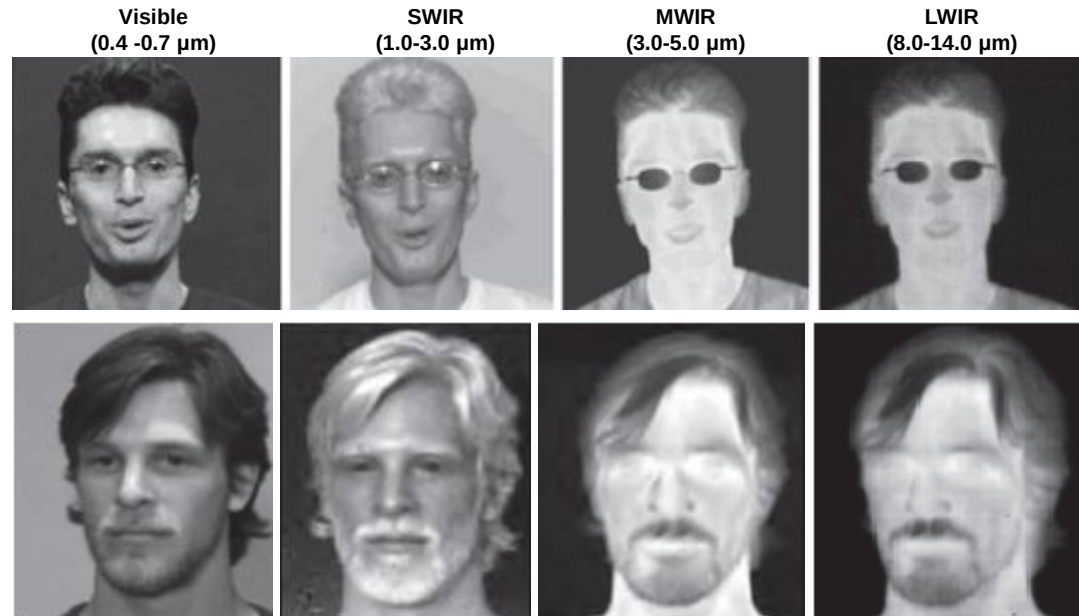
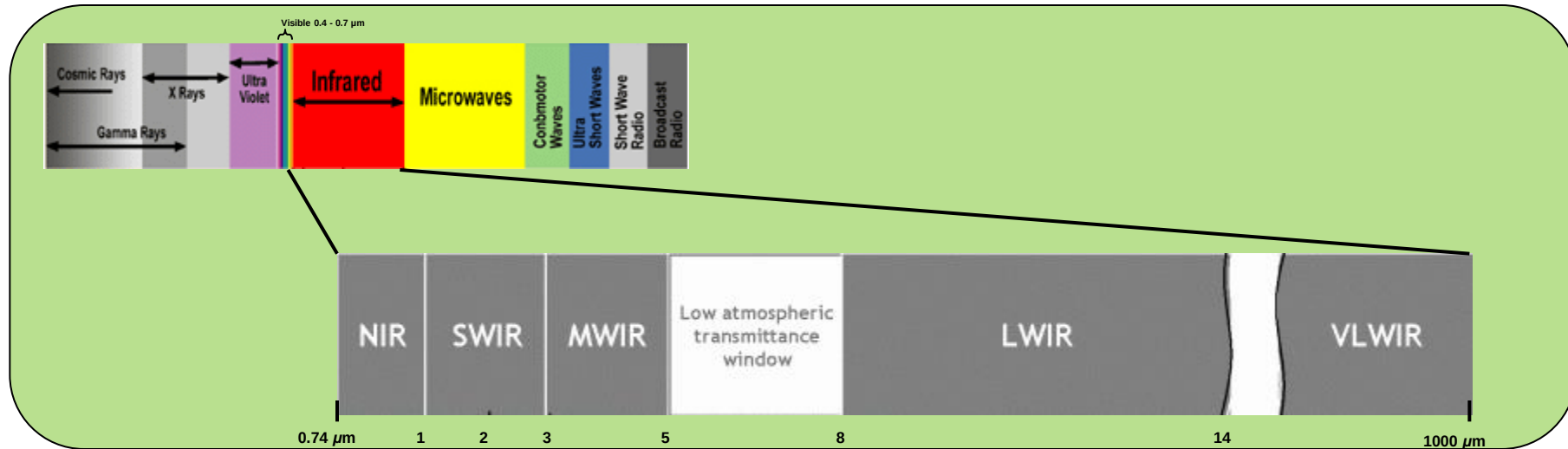
ROC Curves Look-alike Verification

- The Look Alike face database consists of images pertaining to 50 well known personalities from either western, eastern, or Asian origins. Each celebrity has five genuine images and five look-alike images. Each image is of size 220×187 .
- All images re resized to 128×78 .



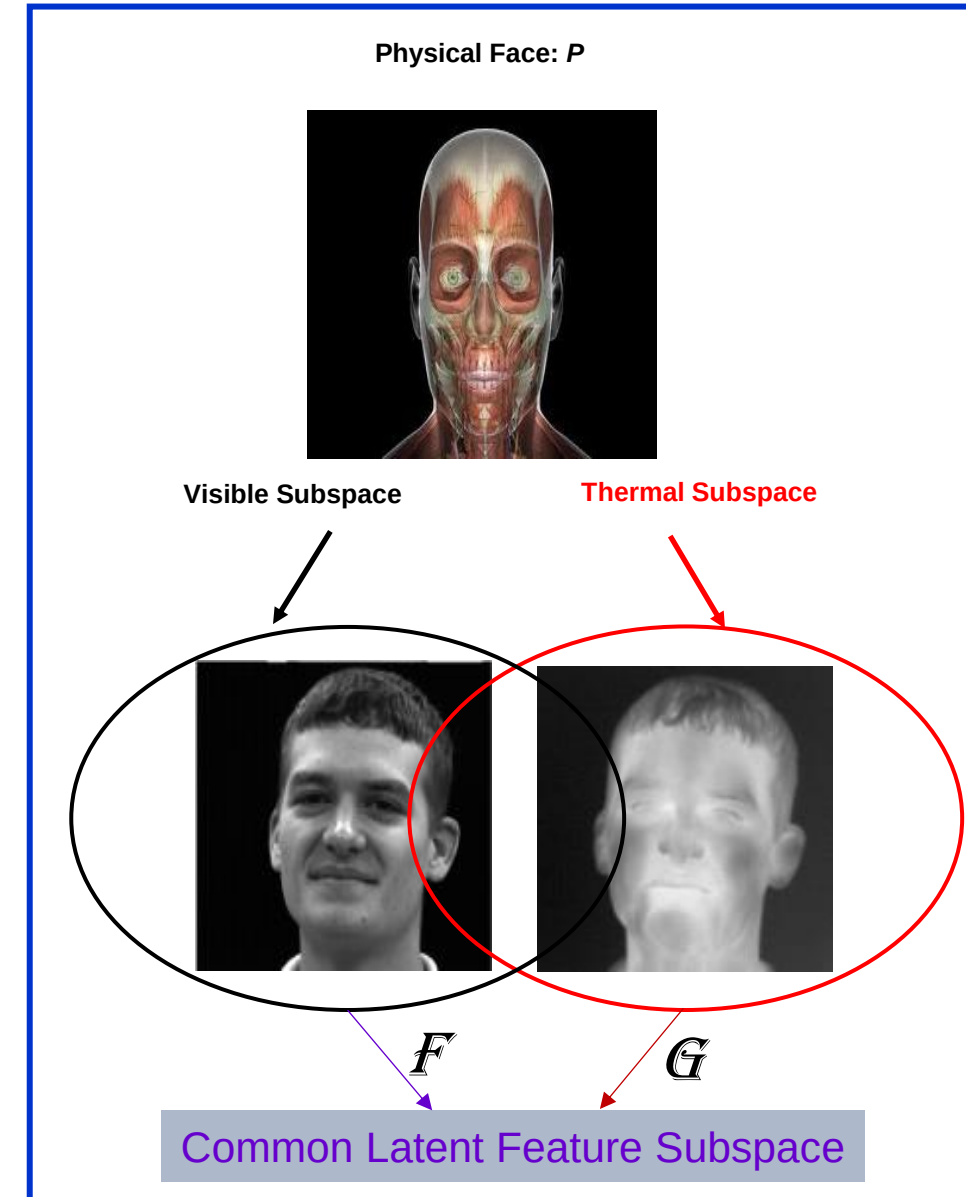
Transfer learning performance on Look Alike dataset

Facial Signatures on Different Spectrum



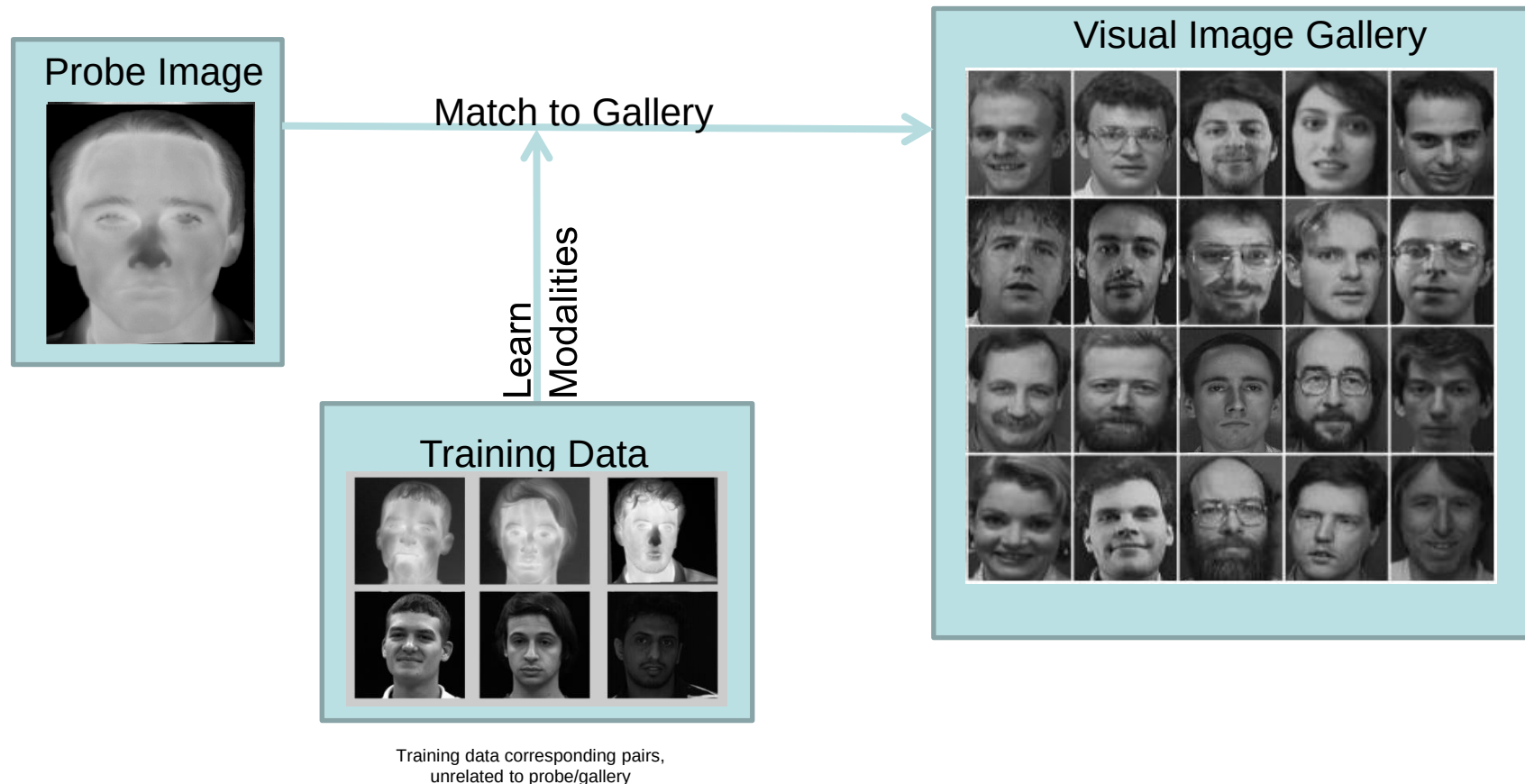
Cross-Spectrum Face Recognition

- **Significance:**
 - Adversaries conduct many activities under cover of darkness, when surveillance with visible imagers is not feasible.
 - Thermal imaging is a maturing technology that has significant potential for face recognition.
 - Research will provide an enhanced nighttime face recognition capability, enabling identification from existing visible spectrum face databases.
- **Key Technical Challenges**
 - Substantial differences between thermal and visible face signatures due to phenomenology.
 - Finding a coupled nonlinear transformations relating thermal to visible signatures.
 - Limited facial details (i.e., limited pixels on target) at extended ranges for surveillance scenarios.



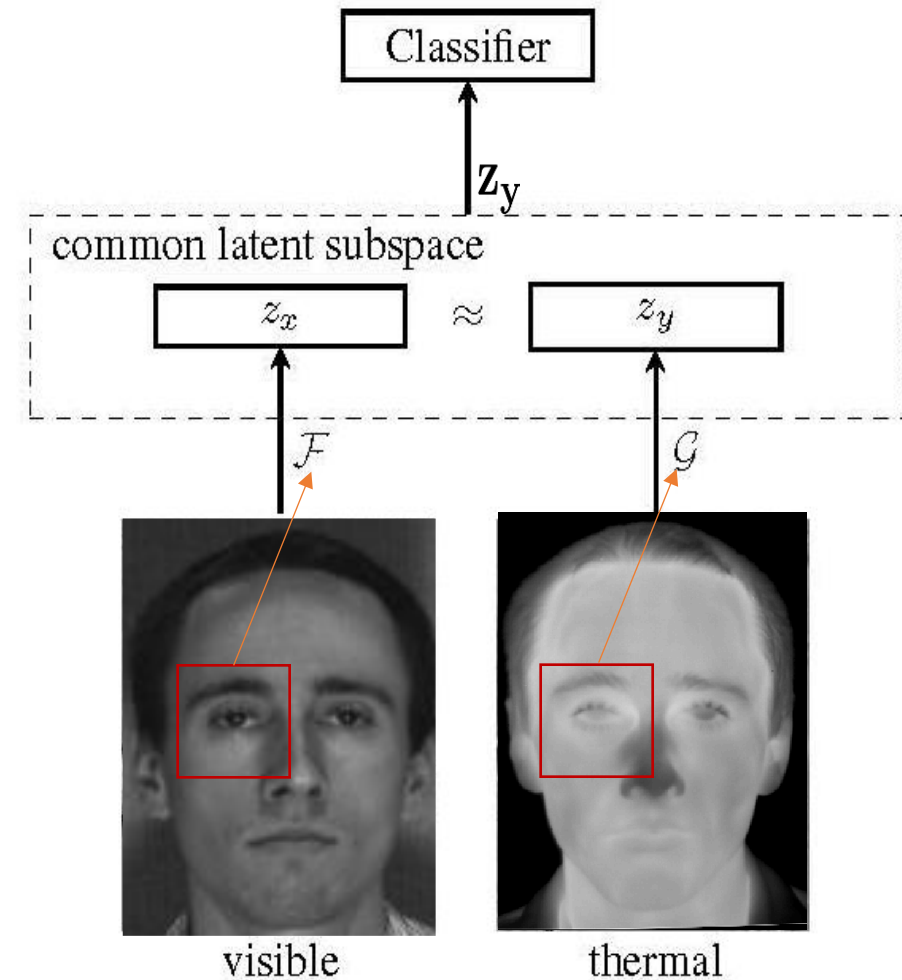
Cross-Spectrum Face Recognition

- Identify person from face via alternative sensing modality.
- We have a training dataset of corresponding thermal-visible pairs to develop our mapping (coupled Neural Network).
- We have **only** the visible images of the probe (test) subjects in a Gallery.



A Common Latent Feature Subspace

- **Objective:** Develop cross-spectrum face recognition approach by obtaining a common latent feature subspace between thermal and visible image signatures
- **Approach:** Find two coupled mappings to transform thermal and visible subspaces to a **common latent feature subspace**
- **Our Methods:**
 1. Coupled Deep Neural Networks (CpDNN)
 2. Coupled dictionary Learning (CpDL)



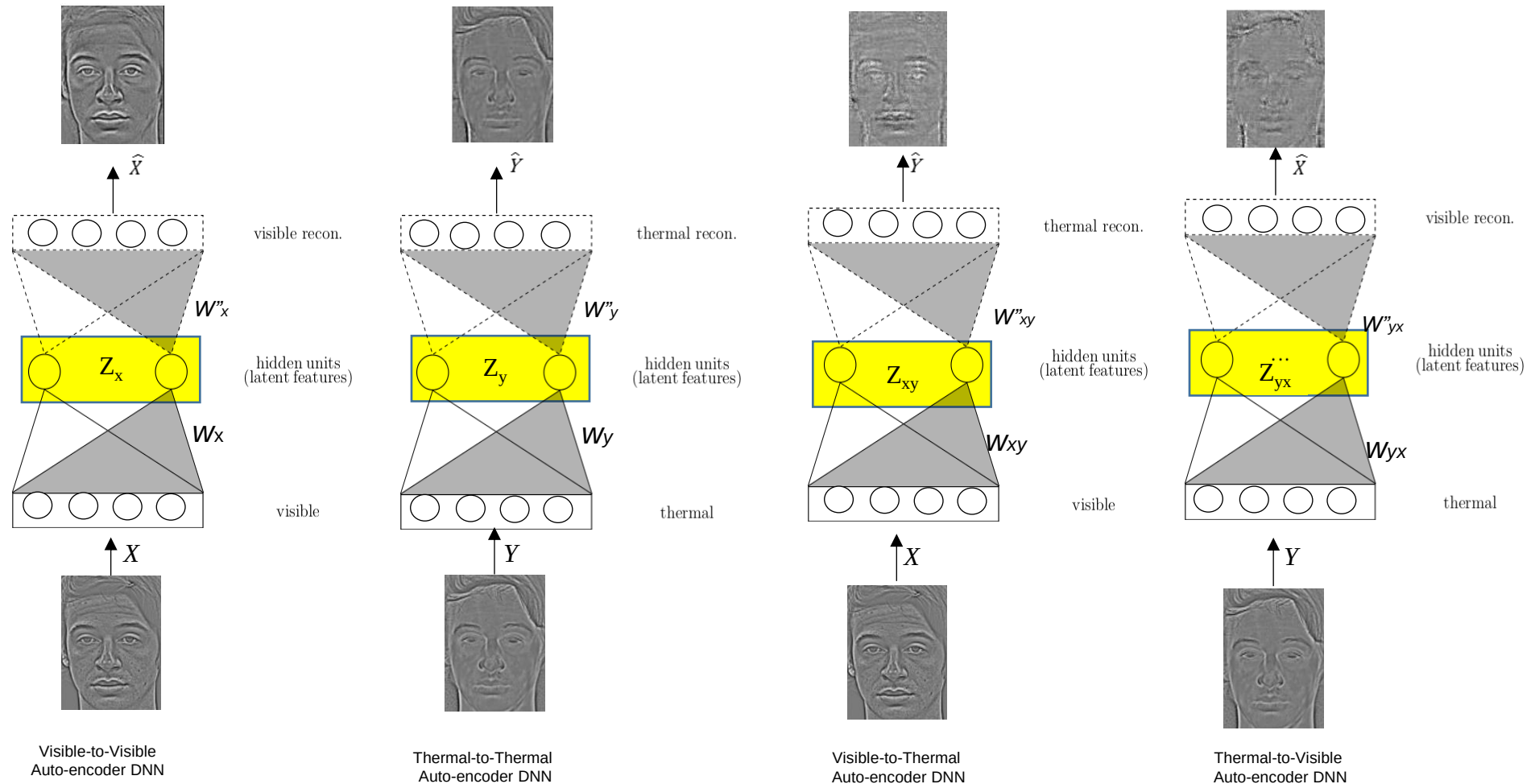
- B. S. Riggan, C. Reale, N.M. Nasrabadi, "Coupled Auto-associative neural networks for heterogeneous face recognition," IEEE Access, vol. 3, pp. 1620-1632, Sept. 2015.
- C. Reale, N. M. Nasrabadi, R. Chellappa "Kernel Task-Driven Coupled Dictionaries for Thermal to Visible Face Recognition," to appear in the IEEE Transactions on Image Processing, 2017.

Autoencoders

Four different neural network autoencoder architectures, output of the hidden units (i.e., Z_X , Z_Y , Z_{XY} , Z_{YX}) are the latent features. For example, Z_{YX} could be used to train a linear SVM classifier.

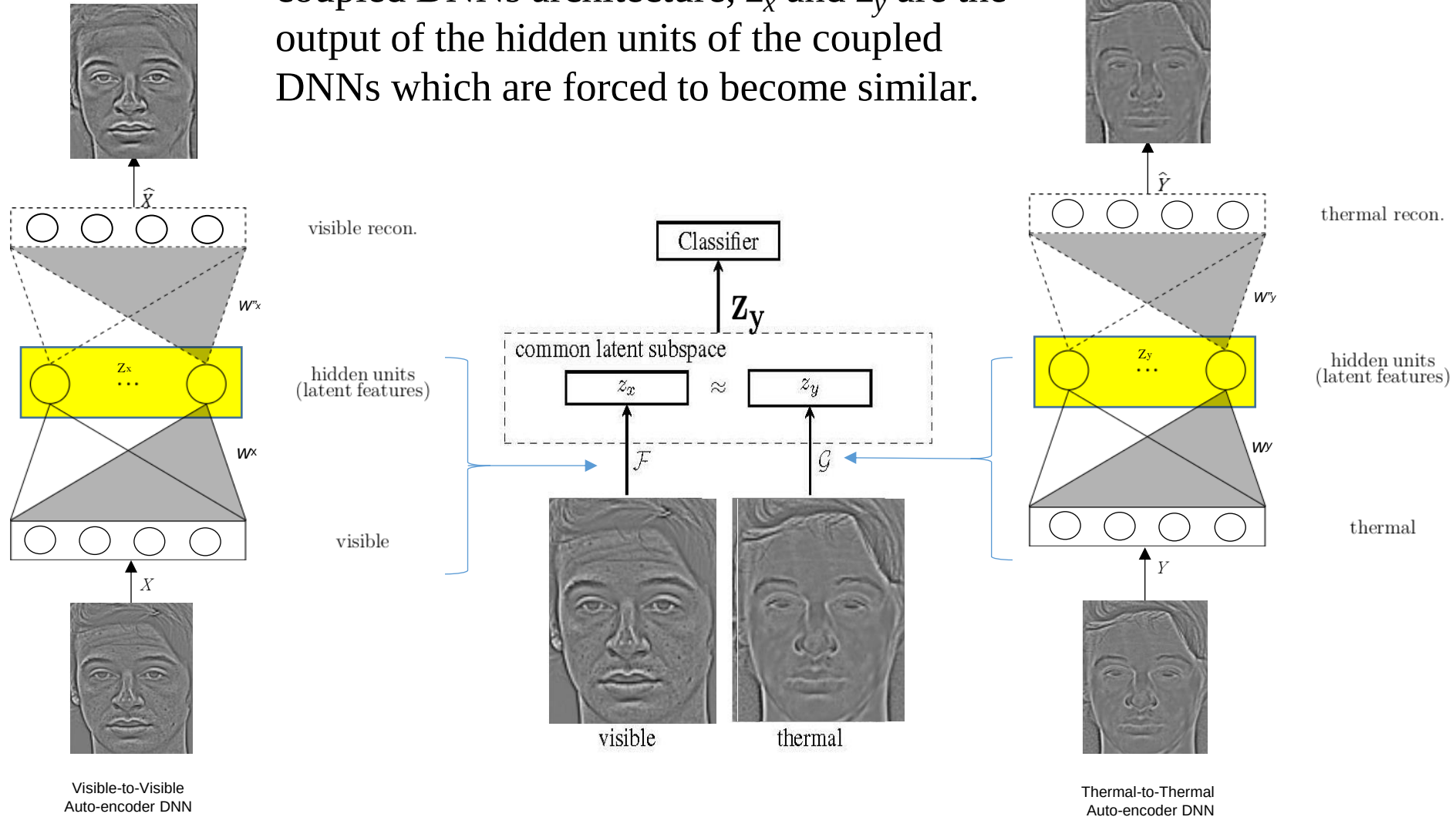
$$R(\Omega_x) = \|X - \hat{X}(\Omega_x)\|_F^2 + \lambda S(\Omega_x) \quad \text{where} \quad \Omega_x = (\omega_x, \omega'_x, b_x, b'_x)$$

Parameters associated with the auto-encoder



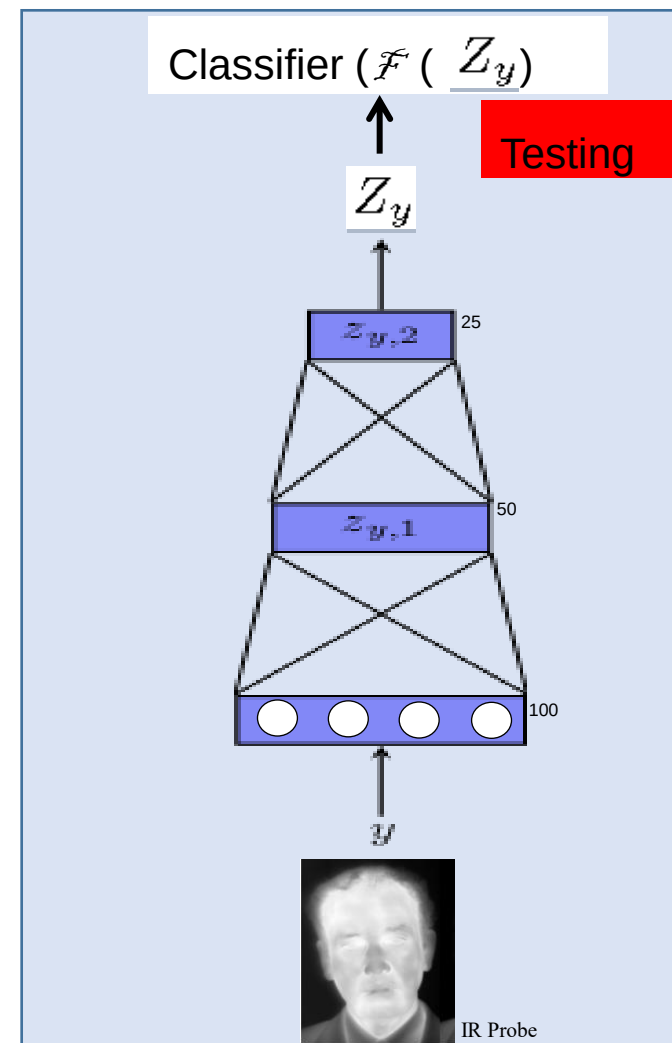
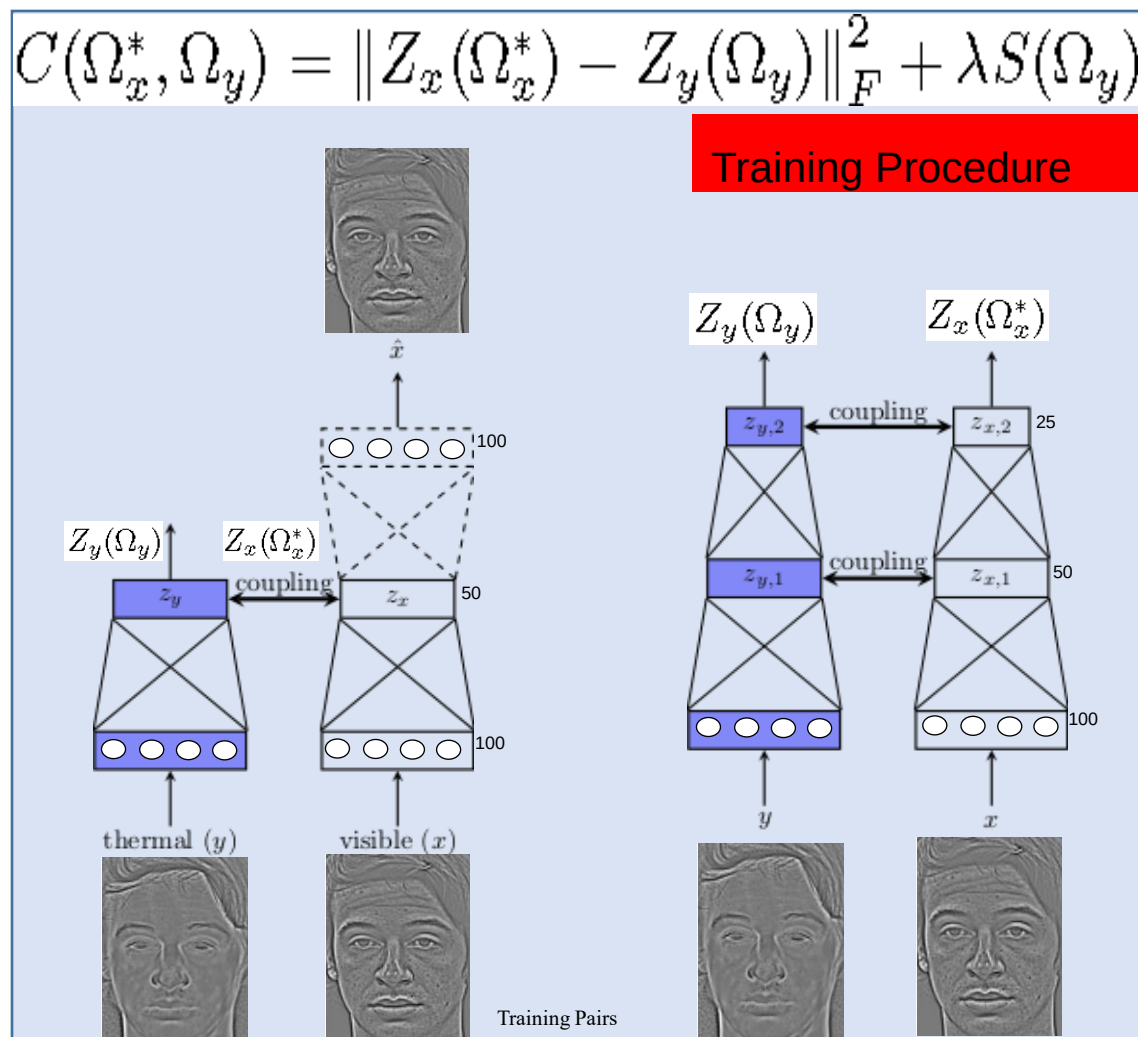
Coupled Deep Neural Networks (DNN) Auto-encoder Architectures

The two mappings are modeled by two coupled DNNs architecture, $\mathbf{z}_x$ and $\mathbf{z}_y$ are the output of the hidden units of the coupled DNNs which are forced to become similar.



Deep Coupled Neural Networks Training

- The architecture of the Cp-DNN encoder-and-decoder is 100-50-25-50-100
- After Cp-DNN is trained on our training dataset, we train a classifier on the visible images of our Gallery subjects and then test it on their corresponding IR probes.



Experimental Results on Two Datasets

- Dataset **WSRI (visible & MWIR)** consisting of 1615 visible-MWIR pairs (110x90 resolution) from 64 different subjects, 50 subjects for training 14 subjects for testing
- Dataset **UND X1 (visible & LWIR)** consisting of 2451 visible-LWIR pairs (110x90 resolution) from 241 different subjects, 159 subjects for training 82 subjects for testing (challenging and a realistic dataset)

Method	WSRI (MWIR vs Visible)	UND X1 (LWIR vs. Visible)
Coupled Dictionary (Reale et al.)	93.1%	50.5%
Coupled Kernel Dictionary (Reale et al.)	95.9%	52.0%
Coupled Deep Neural Network (Riggan et al.)	99.8%	67.0%

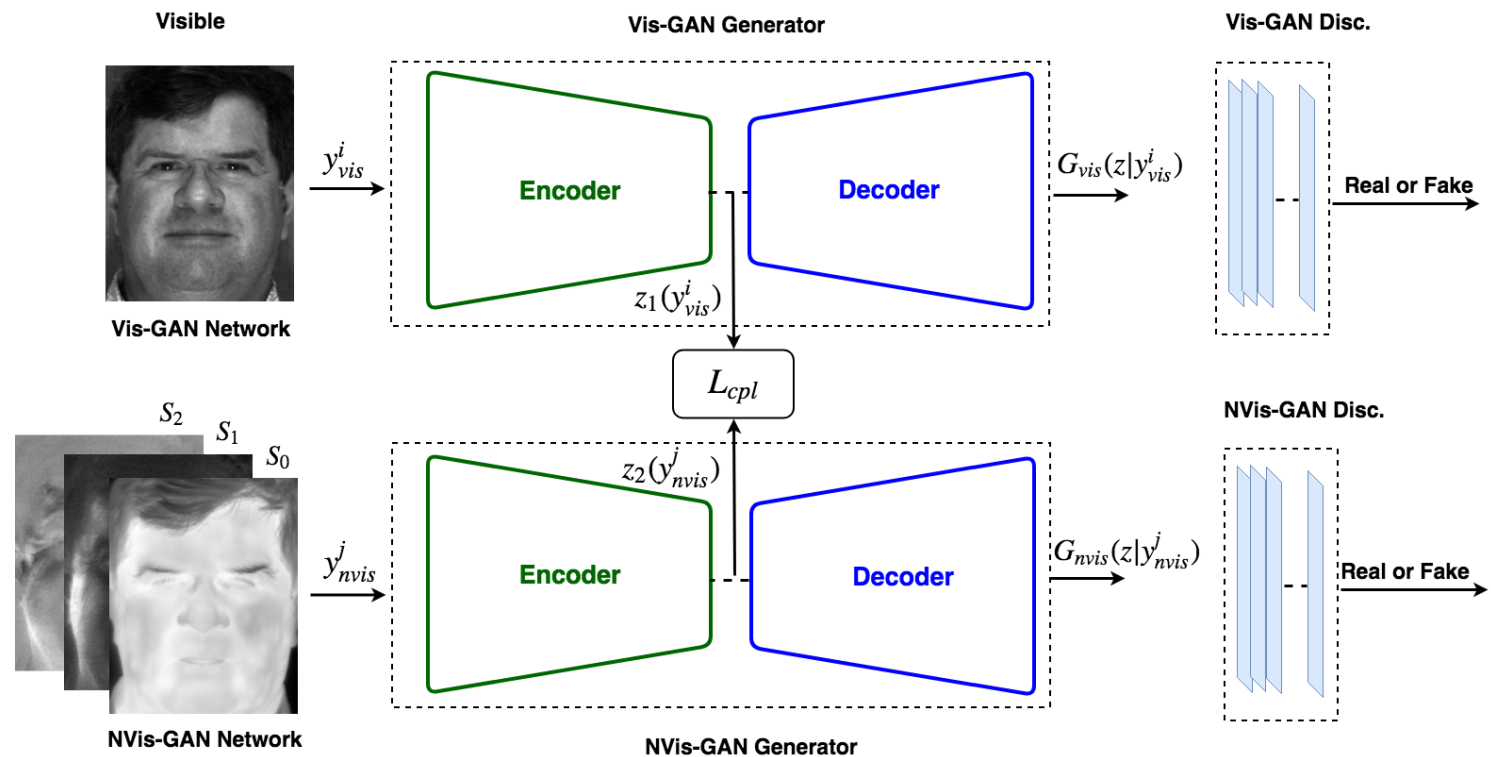
- C. Reale, N. M. Nasrabadi, R. Chellappa “*Kernel Task-Driven Coupled Dictionaries for Thermal to Visible Face Recognition*,” to appear in the IEEE Transactions on Image Processing, 2016.
- B. S. Riggan, C. Reale, N.M. Nasrabadi, “Coupled Auto-associative neural networks for heterogeneous face recognition,” IEEE Access, vol. 3, pp. 1620-1632, Sept. 2015.

Coupled Generative Adversarial Networks CpGAN

Polarimetric thermal-to-visible Face Recognition

- Two generators are coupled with the **contrastive loss function at the bottleneck**.
- Each generator is an autoencoder (e.g., Unet, ResNet, DensNet, etc).
- Discriminators are CNN-based – VGG 19

Polarimetric thermal-to-visible
Face Recognition



[1]SM Iranmanesh, NM Nasrabadi. Coupled Generative Adversarial Networks for Heterogeneous Face Recognition. Image and Vision Computing, vol. 79, Feb. 2020

[2]SM Iranmanesh, A Dabouei, H kazemi, NM Nasrabadi. Deep Cross Polarimetric Thermal-to-visible Face Recognition. ICB 2018.

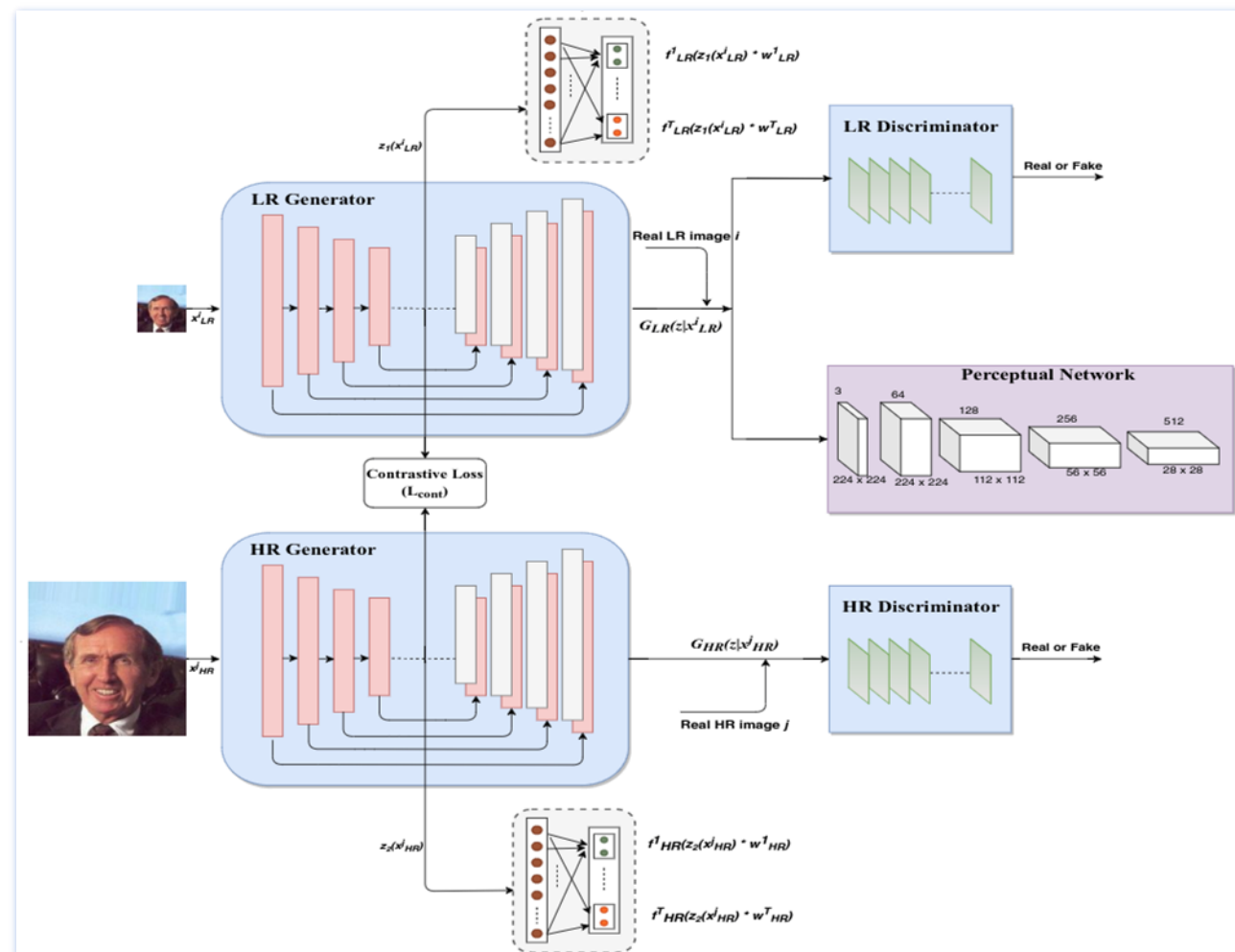
[3]SM Iranmanesh, H kazemi, S Soleymani, A Dabouei, NM Nasrabadi. Deep Sketch-photo Face Recognition Assisted by Facial Attributes. BTAS 2018.

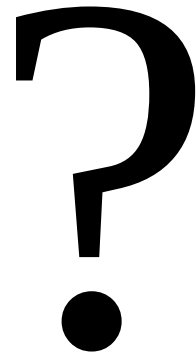
Coupled Generative Adversarial Networks CpGAN

Cross-Resolution Face Recognition

- For both, the **LR and HR sub-networks**, we have used **U-Net network** as the **generator**.
- For both, the **LR and the HR sub-networks**, we have used **patch-based discriminators**.
- Loss functions used to train the overall framework:
 - Contrastive loss function** between the **LR and HR features** retrieved at the **bottleneck** of the **respective generators**.
 - Attribute prediction loss** for the **LR and HR facial images** using the features at the bottleneck of the respective generators.
 - Generative Adversarial Loss** for both the LR and the HR sub-network (GAN)
 - Perceptual Loss only** for the **LR sub-network**
 - L2 reconstruction loss** for both LR and HR sub-networks

V. Talreja, F. Taherkhani, N. M. Nasrabadi, M. Valenti, "Attribute-Guided Coupled GAN for Cross-Resolution Face Recognition," *The Tenth IEEE International Conference on Biometrics: Theory, Applications, and Systems* (BTAS 2019), Sept. 23, 2019, Tampa, FL.





Thank You

Discriminative Siamese Network for Face Verification

- Joplin Hu, Jiwen Lu, Yap-Peng, “Discriminative Deep Metric Learning (DDML) for Face Verification in the Wild,” CVPR 2014
- They modify the contrastive loss

- They minimize $d_G^2(x_i, x_j) = \|G(x_i) - G(x_j)\|_2^2$

with this constrain $y_{ij}(\tau - d_G^2(x_i, x_j)) > 1$

where $y_{ij} = 1$ for genuine pairs and $y_{ij} = -1$ for imposter pairs

$\tau_1 = \tau - 1, \tau_2 = \tau + 1$, respectively.

$d_G^2(x_i, x_j) \leq \tau - 1$ for genuine pair as constrain

$d_G^2(x_i, x_j) \geq \tau + 1$ for imposter pair constrain

- The final optimization is

$$J(W) = \frac{1}{2} \sum_{i,j} f\left(1 - y_{ij}(\tau - d_G^2(x_i, x_j))\right) + \frac{\lambda}{2} \sum_{m=1}^M \left(\|W^{(m)}\|_F^2 + \|b^{(m)}\|_2^2 \right)$$

$f(.)$ logistic function which is the smooth approximation of the Hinge loss function $[z]_+ = \max(0, z)$

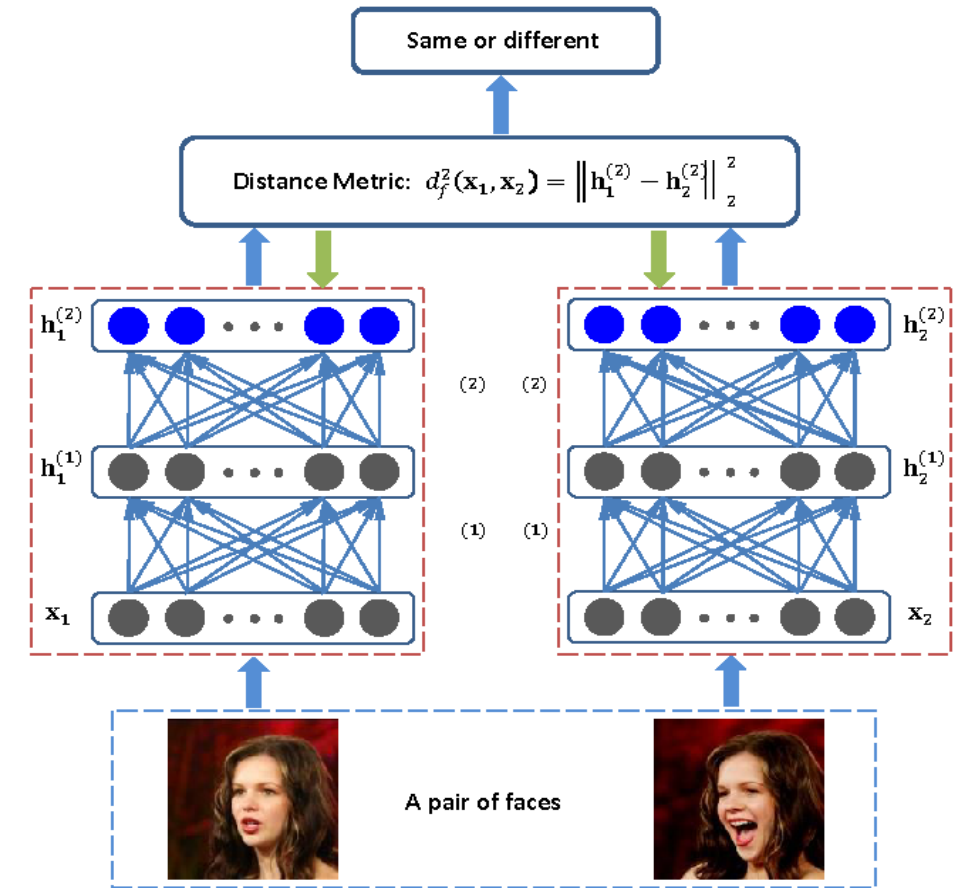


Figure 1. The flowchart of proposed DDML method for face verification. For a given pair of face images x_1 and x_2 , we map them into the same feature subspace as $h_1^{(2)}$ and $h_2^{(2)}$ by using a set of hierarchical nonlinear transformations, where the similarity of their outputs at the most top level is computed and used to determine whether the face pair is from the same person or not.

Discriminative Siamese Network for Face Verification

- They minimize $d_G^2(x_i, x_j) = \|G(x_i) - G(x_j)\|_2^2$ with this constrain $y_{ij}(\tau - d_G^2(x_i, x_j)) > 1$ where $y_{ij} = 1$ for genuine pairs $y_{ij} = -1$ for impostor pairs
 $\tau_1 = \tau - 1, \tau_2 = \tau + 1$, respectively.

- The final optimization is

$$J(W) = \frac{1}{2} \sum_{i,j} f \left(1 - y_{ij}(\tau - d_G^2(x_i, x_j)) \right) + \frac{\lambda}{2} \sum_{m=1}^M \left(\|W^{(m)}\|_F^2 + \|b^{(m)}\|_2^2 \right)$$

$f(.)$ logistic function which is the smooth approximation of the Hinge loss function $[z]_+ = \max(0, z)$

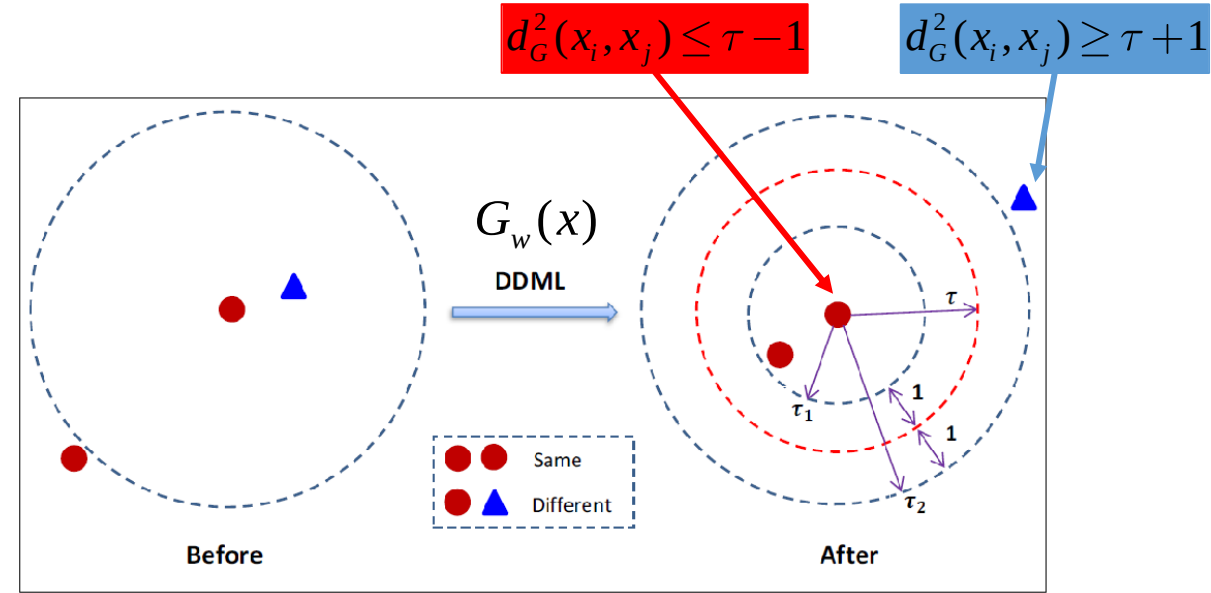


Table 1. Comparison of the mean verification rate and standard error (%) with the shadow metric learning method on the LFW dataset under the image restricted setting.

Feature	DDML	DSML
DSIFT (original)	86.78 $\pm$ 2.09	83.68 $\pm$ 2.06
DSIFT (square root)	87.25 $\pm$ 1.62	84.42 $\pm$ 1.80
LBP (original)	85.47 $\pm$ 1.85	81.88 $\pm$ 1.90
LBP (square root)	87.02 $\pm$ 1.62	84.08 $\pm$ 1.21
SSIFT (original)	86.98 $\pm$ 1.37	84.02 $\pm$ 1.47
SSIFT (square root)	87.83 $\pm$ 0.93	84.52 $\pm$ 1.38
All features	90.68 $\pm$ 1.41	87.45 $\pm$ 1.45